
Embedded EMG-Controlled Gamification System for Rehabilitation of Hand Function after Stroke

Master Thesis:
HST-26-ST-10-10402

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Abstract:

Background & Aim: Stroke is a leading cause of long-term disability, frequently resulting in upper limb motor impairments that limit independence in activities of daily living. To improve activities of daily living performance, intensive repetition of task-oriented motor learning was reported as most effective, while serious games have emerged as promising tools to enhance patient engagement and motivation. EMG-controlled rehabilitation systems have demonstrated improved adherence and hand function, though challenges with usability and setup remain. To address these barriers, the NeuraLoop, a wearable high-density surface EMG human-machine interface, eliminates manual electrode placement and enables real-time muscle activity recording from the forearm, being a promising platform for intuitive and portable stroke rehabilitation systems.

This work aimed to develop a microcontroller-based EMG-controlled gamification system using the NeuraLoop for the rehabilitation of hand function after stroke.

Methods & Materials: Qualitative interview with a therapist ensured that system design aligned with the functional needs and cognitive abilities of stroke patients. The EMG-based gesture detection used a threshold-based algorithm, implemented on the microcontroller of the NeuraLoop device. Detected gestures were mapped to rocket actions in the game.

Results: Microcontroller gesture classification and expected state transitions were fully aligned (accuracy = 100%). The gesture classification was overall reliable (accuracy = 73%), and the mean reaction time from the gesture completion ($-6.36(\pm 194.51)ms$) fell within 300 ms.

Conclusion: This work demonstrated the feasibility of a microcontroller-based EMG-controlled system for interactive rehabilitation gaming.



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Titel:

Mikrocontrollerbaseret EMG-styret
Spilsystem til Rehabilitering af Hånd-
funktion efter Apopleksi

Projekt:

Speciale

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Deltagere:

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Resumé:

Baggrund & Formål: Apopleksi er årsag til langvarig funktionsnedsættelse og medfører ofte motoriske funktionsnedsættelser i overekstremiteten, som begrænser selvstændighed i dagligdags aktiviteter. Intensiv gentagelse af opgaveorienteret motorisk læring er rapporteret til at forbedrer udførelsen af dagligdagsaktiviteter. Serious games vist sig som lovende redskaber til at øge patienters engagement og motivation. EMG-styrede rehabiliteringssystemer har vist forbedret træningsvedholdelse og håndfunktion, men der oplevet udfordringer relateret til brugervenlighed og opsætning. NeuraLoop er et bærbart system baseret på high-density overflade-EMG, som eliminerer behovet for manuel placering af elektroder og muliggør realtidsmålinger af muskelaktivitet fra underarmen. Systemet er en lovende platform for intuitive og mobile rehabiliteringsløsninger til personer med apopleksi. Formålet med dette arbejde var at udvikle et mikrocontrollerbaseret EMG-styret gamification-system ved hjælp af NeuraLoop til rehabilitering af håndfunktion efter apopleksi.

Metoder & Materialer: Et kvalitativt interview med en terapeut blev gennemført for at sikre, at systemdesignet stemte overens med de funktionelle behov og kognitive forudsætninger hos personer med apopleksi. Den EMG-baserede håndbevægelsesdetektion anvendte en tærskelbaseret algoritme implementeret på mikrocontrolleren i NeuraLoop. De registrerede håndbevægelser blev omsat til raket handlinger i spillet.

Resultater: Klassifikationen af håndbevægelser på mikrocontrolleren og de forventede tilstandsovergange var fuldstændigt overensstemmende (accuracy = 100 %). Håndbevægelsesklassifikationen var generelt pålidelig (accuracy = 73 %), og den gennemsnitlige reaktionstid fra afslutningen af en håndbevægelse (-6.36 ± 194.51 ms) lå inden for 300 ms.

Konklusion: Dette arbejde demonstrerede gennemførligheden af et mikrocontrollerbaseret EMG-styret system til interaktiv rehabilitering gennem spilbaseret træning.

Preface

This project is a Master's thesis in Biomedical Engineering and Informatics at Aalborg University, written by group 10402. The project period was from the 1st of February 2026 to the 31st of May 2026. The project was allocated 30 ECTS points and has been developed under guidance of supervisor Erika G. Spaich, Associate Professor, Department of Health Science and Technology at Aalborg University.

Reading Guide

The following report includes a scientific article and associated worksheets. The worksheets are supplementary material and contain chapters concerning different parts of the project. Appendix A is the problem analysis, where the problems associated with upper limb stroke rehabilitation are presented. Appendix B outlines the system design, encompassing requirements, user scenarios, mock-ups, and system interactions. Appendix C details the system testing protocols. Appendix D contains a structured literature review of EMG-based systems integrated with game in hand rehabilitation post-stroke, while Appendix E presents the expert interviews, detailing the analysis and key findings.

Reference Method

The reference method used in the project is the Vancouver method. The reference will appear as a number in square brackets [X]. The number in the square brackets will correspond to the source reference in the bibliography, which is arranged chronologically. If the reference is positioned before the end of a sentence, it means that the reference belongs to that specific sentence. If the reference is referred to as study by XX et al. the year of the reference is defined in round brackets (XXXX).

Use of Generative AI

Generative AI has been used as an assistive tool where ChatGPT and Claude as primary models. The models assisted in code implementation and with grammar and language refinement.



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Abstract

Background & Aim: Stroke is a leading cause of long-term disability, frequently resulting in upper limb motor impairments that limit independence in activities of daily living. To improve activities of daily living performance, intensive repetition of task-oriented motor learning was reported as most effective, while serious games have emerged as promising tools to enhance patient engagement and motivation. EMG-controlled rehabilitation systems have demonstrated improved adherence and hand function, though challenges with usability and setup remain. To address these barriers, the NeuraLoop, a wearable high-density surface EMG human-machine interface, eliminates manual electrode placement and enables real-time muscle activity recording from the forearm, being a promising platform for intuitive and portable stroke rehabilitation systems.

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INTRODUCTION

Cerebrovascular accident (CVA), or stroke, is a neurological disorder caused by disrupted cerebral blood flow [1, 2]. This disruption may result from vascular occlusion called ischemic stroke or from bleeding caused by rupture of cerebral blood vessels, called hemorrhagic stroke, both leading to neuronal injury and cell death through impaired oxygen delivery, increased intracranial pressure, inflammation, and edema [1, 2]. The Global Burden of Disease Study 2021 reported 11.9 mil-

lion new stroke cases worldwide, with 93.8 million people living with stroke, making it the fourth leading cause of disability-adjusted life years globally. Furthermore, it reported ischemic stroke as the most common, with 65% globally [3]. Furthermore, stroke incidence increases with age, with the risk doubling after age 55 for both men and women [1].

Most stroke patients survive the acute phase of stroke, but long-term neurological consequences represent the greatest impact on functional health and quality of life [4]. Stroke may result in neurological and neurophysiologi-

cal complications that affect cognitive, sensory, and motor function [5]. Approximately 35–40% of stroke survivors experience limitations in activities of daily living (ADL) six months after stroke onset, with motor impairment being among the most common deficits [4, 5].

Motor impairment following stroke is often associated with hemiparesis, characterized by weakness on the side of the body contralateral to the lesion, defined as a reduced ability to generate normal muscle strength [6]. These impairments can contribute to abnormal posture, altered reflexes, and loss of voluntary motor control [6]. Upper limb impairments include reduced motor control, sensory deficits, and proprioceptive loss [7]. These impairments result in difficulties performing ADL, including grasping and reaching for objects [8, 9], which affect basic functions such as drinking and dressing [9].

In addition to physical impairments, stroke frequently results in cognitive deficits. A review by Cramer et al. (2023) identified attention deficits, loss of executive function, memory disorders, neglect, aphasia, and apraxia as common cognitive syndromes that affect functional independence and task performance. These deficits may impair concentration, information processing, planning, multitasking, communication, and spatial awareness [10]. As cognitive deficits significantly affect post-stroke patients, their cognitive abilities play a central role in rehabilitation [5, 4].

Upper limb rehabilitation is typically delivered by physiotherapists and occupational therapists during hospitalization and early home-based training [7]. The rehabilitation process involves assessment of the level of functioning, both cognitive and physical, including upper limb function, goal setting, targeted interventions, and continuous evaluation [4]. This involves multidisciplinary teams of medical, nursing, therapy, and social work staff to create individual rehabilitation courses, which include providing intensive, task-specific training to improve functional independence through repeated practice [4]. For rehabilitation after stroke, occupational therapy is often used to improve upper limb function and support independence in ADL through observation of activity limitations and the use of individu-

ally tailored and graded tasks [5, 11]. Effective approaches include mirror therapy, motor imagery and task-oriented training, where the repetition of task-specific motor learning was reported as the most effective [5, 12]. Furthermore, constraint-induced movement therapy (CIMT) has also been shown to be effective for improving paretic upper limb function [13, 14, 15]. This approach forces the use of the affected arm by restraining the unaffected arm during exercise sessions or usual ADL [13, 14, 15]. To support task-specific rehabilitation of hand and wrist function, understanding the neuromuscular impairments underlying upper limb motor control is important.

Coordinated activation between flexor and extensor muscles is essential for controlled wrist and hand movements [16]. After a stroke, muscle activation patterns are often disrupted, resulting in reduced wrist control and impaired motor control [17]. Increased flexor muscle activity and insufficient extensor activation result in the characteristic flexed wrist and clenched hand posture frequently observed in stroke patients [18]. The disrupted activation patterns affect grasping ability and performance of ADL tasks, making restoration of balanced muscle activation an important focus in upper limb rehabilitation [17, 18]. These altered activation patterns can be assessed using electromyography (EMG), recording the electrical activity generated during muscle contraction [19]. In upper limb stroke rehabilitation, surface EMG (sEMG) is widely used to evaluate muscle activation and co-activation patterns in the forearm flexors and extensors, providing insight into impaired motor control and movement coordination [20, 21, 22]. Furthermore, high-density surface EMG (HD-sEMG) enables more detailed mapping of neuromuscular activity, with potential to identify altered coordination patterns even in patients with limited hand function [23, 22]. Maintaining patient motivation and engagement is an important aspect of upper limb rehabilitation after stroke, as recovery often requires intensive and repetitive training over long periods of time [7, 4, 12, 24]. To improve adherence and make rehabilitation exercises more engaging, serious games and virtual reality (VR) have therefore become an important focus in neurorehabilita-

tion research [25, 26].

EMG-controlled serious games and virtual reality (VR) systems have demonstrated high levels of engagement and motivation for stroke rehabilitation [26, 25, 27, 28]. In these systems, sEMG signals from forearm muscles are used as a human-machine interface, where muscle activation is translated into control commands for computer games or virtual environments [26, 25, 27, 28]. This enables stroke patients to perform task-specific movements while receiving immediate visual feedback, supporting motor learning and active participation in rehabilitation exercises [25, 26, 29, 30]. Ito et al. (2023, 2024) further reported increased engagement with repetitive tasks compared to conventional exercises.

Several studies have demonstrated improved participant adherence, hand function, and grip strength using EMG-controlled rehabilitation games [31, 32, 33, 34, 35]. Marin-Pardo et al. (2022, 2023, 2024) evaluated Tele-REINVENT, a low-cost home-based sEMG-controlled system that controlled adaptive computer games for stroke rehabilitation. Similarly, Amin et al. (2025) investigated sEMG-controlled VR-based games, while Seo et al. (2022) trained chronic stroke patients using an sEMG-controlled game to retrain hand muscle activation.

Despite the benefits, practical barriers remain for less therapist-supervised sEMG-controlled rehabilitation systems. Munoz-Novoa et al. (2024) and Meyers et al. (2024) identified electrode placement, setup difficulties, and system complexity as challenges that may reduce the system adoption among users. In addition to hardware usability, intuitive game design has been highlighted as an important factor for maintaining patient engagement and adherence in EMG-based rehabilitation systems [36, 30, 24, 26]. Mejía et al. (2019) emphasized the importance of an intuitive interaction design, while Ito et al. (2021) demonstrated increased usability and enjoyment using a simple and intuitive EMG-controlled game design for upper limb rehabilitation on stroke patients. Similarly, Ali et al. (2024) [37] reported that intuitive gaming elements can improve motivation, engagement, and adherence, thereby supporting more intensive upper limb

training after stroke. To address the setup and usability barriers, Meyers et al. (2024) and Murciego et al. (2022, 2023, 2025) both developed wearable forearm HD-sEMG sleeves that span electrodes across the entire forearm, eliminating the need for manual electrode placement on specific muscles. Meyers et al. (2024) further demonstrated that the sleeve could be used by the user without assistance.

The NeuraLoop presented by Murciego et al. (2022, 2023, 2025) is a wearable HD-sEMG human-machine-interface designed for real-time interaction and rehabilitation applications. The system records muscle activity from the forearm using a wearable HD-sEMG sleeve and enables closed-loop interactions by recording the user’s muscle activity while simultaneously delivering real-time haptic feedback. EMG data are streamed to an external computer for signal processing, including gesture detection and classification. However, Murciego et al. (2025) highlights potential for fully embedded real-time processing directly on the NeuraLoop device, supporting usability and portable rehabilitation systems.

To enable intuitive interaction between the HD-sEMG sleeve and the rehabilitation game, reliable translation of muscle activation into control commands is required. This creates a need for gesture recognition methods to translate the EMG signal into specific control commands.

Gesture recognition on EMG signals has been widely studied using both threshold-based and machine learning approaches [38, 39]. Among the machine learning approaches, linear discriminant analysis (LDA) and support vector machines (SVM) are widely used for EMG-based gesture recognition due to their high classification accuracy and suitability for real-time applications [40, 41, 42]. Murciego et al. (2022, 2023) demonstrated classification accuracies above 90% using LDA for hand gesture classification, while Murciego et al. (2025) achieved precision above 90% and recall above 95% using an SVM classifier for real-time classification of four thumb micro-gestures. Previous studies also demonstrated that both LDA and SVM algorithms can be implemented efficiently on microcontroller unit (MCU) for embedded EMG-control systems (accuracy >

90%), with the purpose of controlling prosthetic arm prosthesis [43, 44]. Furthermore, a review by Castruita-Lopez et al. (2025)[45] highlights the low computational cost and fast performance of LDA- and SVM-based gesture recognition methods for MCU applications.

Besides machine learning approaches, simpler threshold-based methods are widely used for EMG control systems. The static threshold method, first proposed by Di Fabio (1987)[46], remains one of the most widely used approaches for detecting muscle activation onset in EMG signals. In this method, the amplitude of the processed EMG envelope is compared against a predefined threshold to identify active muscle segments corresponding to intended gestures. Due to its low computational complexity and ease of implementation, threshold-based activation detection has been extensively applied in more recent prosthetic and rehabilitation systems, where static thresholding methods provide reliable intuitive control for basic prosthetic and rehabilitation tasks [38, 47, 48, 49, 50].

Despite the simplicity and robustness of threshold-based methods, EMG control systems are often affected by signal noise, electrode displacement [39], and unintended state switching [51, 39]. To address these issues, hysteresis mechanisms have been introduced by applying activation and deactivation thresholds, which improve stability by reducing small, rapid oscillations near the decision boundaries [51]. The continued use of threshold-based control aligns with the findings of previous studies [38, 47, 48, 49, 50], which emphasize that advanced pattern recognition algorithms have shown promising results in laboratory performance, but many systems still prefer simpler threshold-based control approaches due to their reliability, lower training requirements, and ease of implementation in real-world applications [39]. This highlights the practical relevance of threshold-based EMG control for embedded systems where computation efficiency and robustness are critical design considerations.

Although previous studies have demonstrated the potential of EMG-controlled rehabilitation systems, challenges remain in developing systems that combine intuitive interac-

tion, embedded real-time processing, and usability for stroke rehabilitation. Existing game-based hand rehabilitation systems often rely on external computers for signal processing or require complex setup procedures, limiting their practical applicability outside laboratory settings.

This work aims to develop a MCU-based EMG-controlled gamification system using the NeuraLoop for the rehabilitation of hand function after stroke.

SOLUTION

To support hand rehabilitation after stroke, this work proposes an embedded MCU-based gamification system designed to enhance patient engagement during rehabilitation exercises. The primary objective of the system is to increase adherence to repetitive motor training by integrating real-time feedback during hand rehabilitation post-stroke. The proposed embedded rehabilitation system consists of the NeuraLoop device, which integrates an MCU-based signal acquisition and processing with an HD-sEMG sleeve for recording forearm muscle activity. A gesture detection algorithm implemented on the MCU processes the EMG signals in real-time to identify user hand gestures. The detected gestures are subsequently mapped to corresponding in-game actions, allowing the user to control a rocket and thereby enabling interactive gamified rehabilitation exercises and real-time visual feedback of movement performance during the training session.

The proposed solution is structured into three main phases: system design, implementation, and testing, which assess functionality in accordance with the system design and requirements specification.

NEURALOOP SPECIFICATIONS AND SETUP

NeuraLoop specifications

The NeuraLoop, developed by Murciego et al. (2022, 2023, 2025) is an intuitive, high-bandwidth bidirectional human-machine interface. It enables closed-loop interaction by recording HD-sEMG while simultaneously de-

living real-time haptic feedback. The NeuraLoop system consists of a flexible dry electrode array printed into a textile forearm band for monopolar EMG recordings, connected to a compact electronic device for both signal recording and stimulation (Figure 1a). In this work, the stimulation functionality was not used. The textile sleeve used in this work has a 32-pad electrode array for EMG recording (Figure 1b).

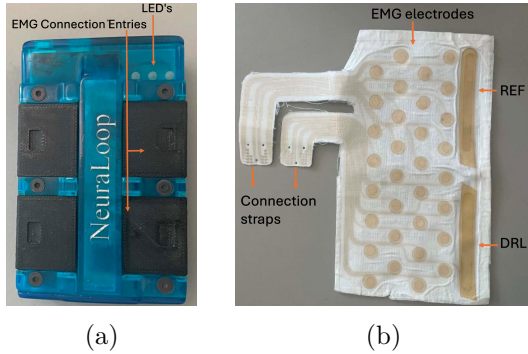


Figure 1: (a) The compact electronic NeuraLoop device with 2 connection entries for HD-sEMG sleeve connection, and (b) HD-sEMG sleeve with 32 sEMG electrodes, a reference (REF) electrode, a driven right leg (DRL) electrode, and two connection straps to connect to the NeuraLoop device.

Besides the recording module, the electronic system consists of a power supply module, a USB connection for charging, and an MCU. The MCU is a 32-bit ARM MCU (STM32H743VIT6, ST Microelectronics). Furthermore, communication through the WiFi module occurred via UART, enabling the MCU to transmit EMG data to a host computer using a Transmission Control Protocol (TCP) connection. STM32CubeIDE was used to update and create firmware files on the MCU. Furthermore, the EMG signals were sampled at 1000 Hz and transmitted via WiFi at a baud rate of 5,000,000 bit/s, corresponding to 625,000 byte/s, enabling high-speed transmission of EMG data to the receiving computer.

Attachment of the HD-sEMG sleeve and the NeuraLoop

During the recording session, the participants sat comfortably on a chair in front of a computer screen. The skin was cleaned using abra-

sive paste (Everi, Spes Medica, Genoa, Italy) before placing the HD-sEMG sleeve (Figure 2).



Figure 2: NeuraLoop and HD-sEMG sleeve attached to the right forearm.

The HD-sEMG sleeve was positioned approximately 3 cm below the wrist so that the reference electrode was located distal to the primary muscle bellies of the extrinsic forearm muscles. This reduced the risk of the reference electrode capturing myoelectric activity from the extensor muscles, which could otherwise interfere with the differential EMG recording. Depending on the participant's forearm size, the HD-sEMG sleeve either slightly overlapped or left a small gap at the connecting edge. The connection edge was therefore aligned with the ulna on the underside of the forearm. To minimize movement artifacts, the HD-sEMG sleeve was taped tightly to the forearm, and a compression sleeve was further applied to ensure tight electrode contact. Finally, the NeuraLoop device was connected to the HD-sEMG sleeve and secured with an elastic bandage to prevent movement during the training session (Figure 2).

SYSTEM DESIGN

Functional specifications were designed based on a qualitative interview with an experienced therapist at a rehabilitation center, ensuring alignment with the functional needs and cognitive abilities of the users (Appendix E). Three key considerations emerged from the interview. First, exercises targeting reaching and grasping, ADL, edge positioning, open/close hand movements, and the activation of the thumb were identified as highly relevant to upper limb rehabilitation. Second, given the cognitive challenges commonly experienced by stroke pa-

tients, it was deemed essential that training exercises be simple, intuitive, and clearly communicate how each movement must be performed. Third, the system should actively support patient motivation and long-term adherence to training. In continuation of this, the therapist emphasized that patients are guided to take responsibility for their own rehabilitation, where therapists, among other things, guide patients to document their training in their personal training folders. Support for most of these considerations can also be found in the literature, where studies highlight the importance of repetitive training of wrist flexion and extension following stroke [52, 53, 54]. Wrist flexion and extension exercises play a central role in post-stroke upper limb rehabilitation, as targeted training of the wrist flexor and extensor muscles has been shown to improve motor control and muscle activation [55, 53]. Particularly, wrist extension is essential for effective grasping and object manipulation, as it directly influences grip strength and hand function [53].

The system involves two primary actors: the therapist and the patient. The therapist configures the training session by selecting exercise combinations and training dosage and initiates the session. During the training session, the patient attempts to perform different exercise combinations to support the rehabilitation of hand function. The exercise combinations of hand movements listed below:

- Movements from edge position to wrist flexion, and from edge position to wrist extension (EFE).
- Movements from edge position to wrist flexion, from edge position to wrist extension, and from edge position to a closed hand (EFEC).
- Movements from edge position to wrist flexion, from edge position to wrist extension, and from edge position to an open hand (EFEO).
- Movements from edge position to wrist flexion, from edge position to wrist extension, from edge position to a closed hand, and from edge position to an open hand (EFEEO).

The patient then performs the exercises guided by visual feedback and is informed about completed repetitions at the end of the session.

Based on the considerations from the therapist interview and supporting literature, the user requirements are presented in Table 1.

In the implemented system, only a limited set of actions was implemented due to the time constraints of this work. Therefore, only the requirements regarding these functions were evaluated. System requirements related to the MoSCoW [56] must-have user requirements were prioritized, as these were considered essential functionalities of the system. The full

Table 1: *User Requirements prioritized according to MoSCoW.*

ID	Requirement	MoSCoW
UR1	The therapist must have an interface to set up training sessions with training dosage and exercise combinations consisting of edge position, wrist flexion, wrist extension, close and open hand.	Must
UR2	The patient must receive support to train edge position.	Must
UR3	The patient must receive support to train wrist flexion.	Must
UR4	The patient must receive support to train wrist extension.	Must
UR5	The patient must receive support to train closing the hand.	Should
UR6	The patient must receive support to train opening the hand.	Should
UR7	The patient must be informed how to perform edge position, wrist flexion, wrist extension, closing, and opening of the hand to complete the exercises.	Must
UR8	If the patient is inactive for more than 30 seconds, the system must remind the patient.	Could
UR9	The patient must be informed about the number of repetitions at the end of the session.	Could

list of system requirements can be found in Appendix B.

The system requirements not derived from the user requirements, defined essential and operational conditions necessary to fulfill the user requirements. These system requirements were identified through the system design process. The system requirements include an interactive interface, connection to NeuraLoop, and calibration of muscle activity (Table 4). The system requirements derived from the user requirements define how the corresponding user requirements were fulfilled (Table 5). The system’s workflow is presented in the activity diagram (Figure 3) where the implemented system is highlighted with gray boxes. The left side presents the overall system, where the therapist assists during *Calibrate for Muscle Activity Level* to ensure that channel selection and threshold selection proceed optimally before the training session starts. Additionally, the therapist assists with *Open Application* and sets up the training session by first performing *Choose Channels and Thresholds* and afterwards *Connection to NeuraLoop*, *Choose Training Dosage*, and *Choose Movements*. The therapist is responsible for starting the training session, where the patient is responsible for playing the game. The game progresses through different scenarios, while the patient’s hand movements are mapped into rocket actions. When the training dosage is achieved, the training session is completed. See Appendix B for the design description of the remaining system components that were not implemented.

IMPLEMENTED SYSTEM

The system was implemented on the MCU of the NeuraLoop device and utilized EMG signals to detect the user’s hand movements in real-time. Prior to use, a calibration procedure was performed to identify the most informative EMG channels and threshold values. Following calibration, the therapist entered the selected EMG channels and corresponding threshold values into the system. The training setup was then configured by selecting the **Side to Side** exercise, **Right Hand**, and the desired training dosage. The user inter-

acted with the game in which a rocket was controlled through detected hand movements derived from muscle activity measured using EMG. The rocket could be moved to the left or right in order to avoid incoming asteroids. Movement detection was performed using a threshold-based gesture recognition algorithm, which classified muscle activity as either flexion or extension, mapped to leftward or rightward rocket movements. When edge position was detected, the rocket moved forward. During the game, the progress bar in the game showed the training dosage of the training session. When the game ended, the accumulated counts of detected hand movements were displayed to remind the user of documenting the results of the training session.

The implementation of the system requirements is described below.

Connection to NeuraLoop and Game Application:

This implementation consisted of ensuring the establishment of a connection between NeuraLoop and the game application, as well as the implementation of the user interface at the start screen and game screen. The implementation fulfilled the following system requirements: SR0.1, SR0.3, SR0.4, SR0.5, SR0.7, SR0.8, SR1.1, SR1.2, and SR7.1.

The system was developed in Unity and built as a standalone executable file. Since the system integrated Python-based functionality, the build process was automated through a Bash script that compiles the Unity project, packages the required Python dependencies, and generates installation and execution scripts. After the initial one-time installation, the application could be launched directly through the execution script. The application provided a step-by-step connection guide at the start screen (Figure 4) that instructs the user to power on the NeuraLoop, connect to its WiFi network, and press the connect button, though calibration parameters must be configured before the guide was followed. Communication with the NeuraLoop was established through a TCP connection via a Python-based relay script, which configured EMG channels and hysteresis threshold bands and acquired gesture data in real-time; the script extracts gesture labels from incoming packets and forwards

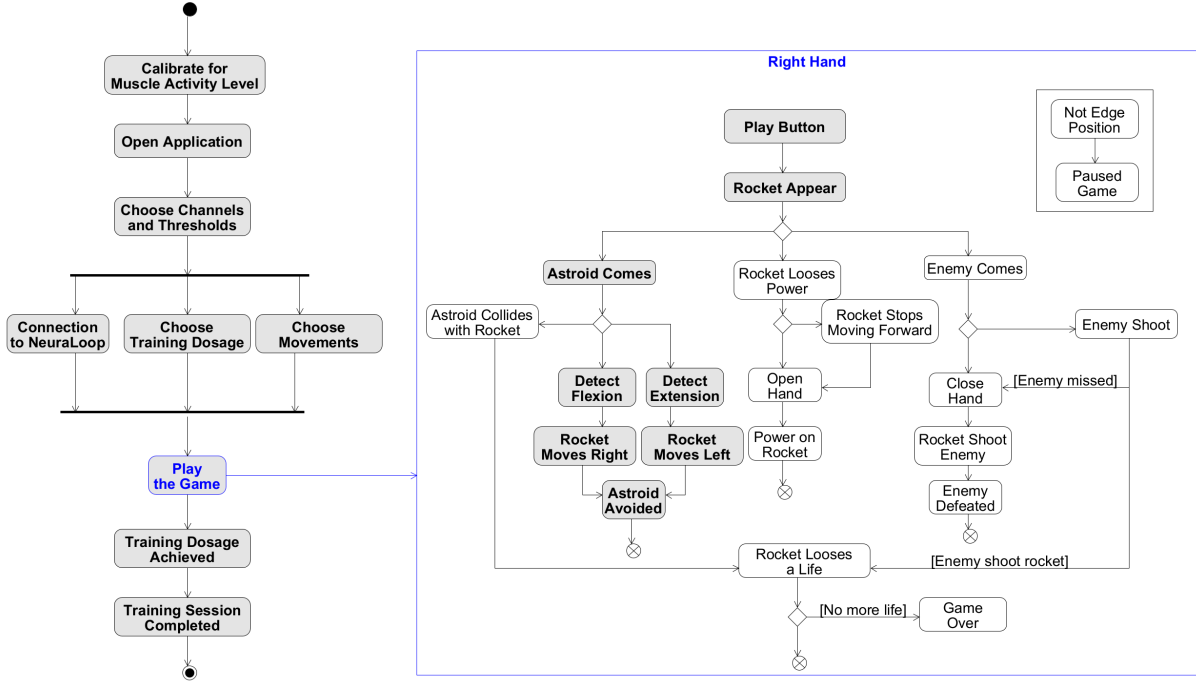


Figure 3: Activity diagram showing the workflow of the system. The arrows represent the direction of the workflow, rounded rectangles represent actions, and diamonds represent decisions with different conditions. On the left side, the overall activities of the system are shown, where the game is shown in more detail, in the blue box on the right side. Gray boxes illustrate the implemented system.

them to the Unity application using the Open Sound Control(OSC) protocol. Throughout this process, the connection status was communicated to the user via a color-coded status field with a corresponding text label, transitioning from dark red when disconnected, to dark orange while connecting, to dark green upon successful connection, and to red if the connection failed.

The start screen further consisted of a drop-down list for exercise selection and a drop-down list for hand selection in the main menu (Figure 4), where the user chooses between the exercise combinations, EFE, EFEC, EFEO, and EFECO, as well as right or left hand. Each exercise combination was mapped into corresponding in-game rocket actions referred to as play modes, as listed below:

- EFE mapped to Side to Side
- EFEC mapped to Side to Side + Shoot
- EFEO mapped to Side to Side + Power
- EFECO mapped to Side to Side + Power + Shoot

The selected mode was stored and applied when the game was loaded. Additionally, an input field was placed for specifying training dosage by a target distance (Figure 4).



Figure 4: Visualization of the game start screen including input fields for the calibration parameters, NeuraLoop connection guide, and training setup with a drop-down list for hand selection, drop-down list for exercise selection, and input field to define the training dosage.

The duration of the training session was adjusted to the specified training dosage. When the play button at the bottom of the start screen was activated, the game screen was

loaded, and the training session started. The completed training dosage was tracked continuously by accumulating the distance travelled based on the elapsed time, displayed in real-time through a progress bar in the game screen (Figure 5). A visualization of the hand movements for edge position, wrist flexion, and wrist extension was displayed at the bottom of the game screen, with the corresponding directional arrow shown below each movement (Figure 5). The training session ended automatically when the target dosage was completed, and the end screen appeared (Figure 12).

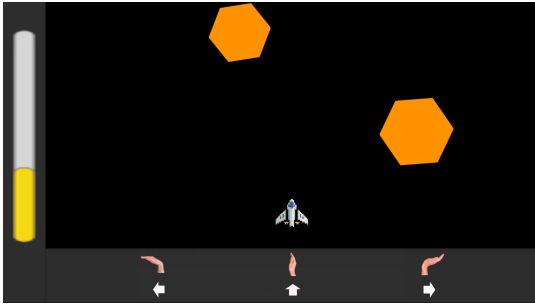


Figure 5: Visualization of the game screen consisting of the rocket with thrust behind it to visualize the rocket’s forward momentum, meanwhile asteroids are spawning from the top, which must be avoided by the rocket. The game screen further consists of a visualization of hand movement at the bottom and a progression bar on the left side.

Calibration:

This implementation of the calibration procedure ensured that the system adapted to the user’s muscle activity before the training session began, fulfilling the system requirement: SR0.6.

Calibration was performed using a MATLAB-based tool that guided the subject through a 45-second protocol of alternating wrist flexion, wrist extension, and rest corresponding to edge position (Figure 6). The subject followed a trapezoid cue, ramping up to maximum voluntary contraction (MVC) at a three-second peak of each movement phase. Channel selection was based on a contrast score that rewarded activation during the target movement while down-weighting co-activation during the antagonist movement. For each EMG channel, a rectified and smoothed EMG envelope was computed. Mean activation

level was then calculated for rest(m_{Rest}), flexion(m_{Flex}), and extension(m_{Ext}) periods, and a small constant was added to avoid zero division (ε) (Equation 1 shown for flexio).

$$c_{Flex} = \frac{m_{Flex}}{m_{Rest} + \varepsilon} \quad (1)$$

To quantify movement specificity, a contrast-score was calculated for each channel (Equation 1 shown for flexion)

$$S_{Flex} = c_{Flex} \cdot \max(c_{Flex} - c_{Ext}, 0) \quad (2)$$

The channels with the three highest scores for flexion and extension were selected. Their envelopes were then averaged to form two composite signals. The difference between these signals was used as the classifier input for movement detection. Entry and exit threshold values were manually selected.

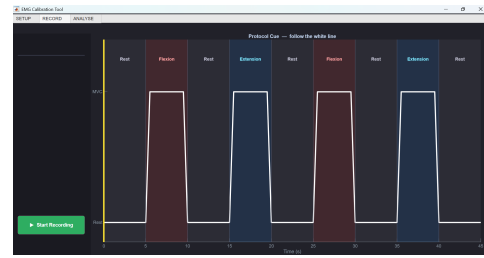


Figure 6: Visualization of the 45-second calibration protocol including a trapezoid cue to demonstrate when the user should perform flexion, extension, and rest corresponding to edge position.

Gesture Recognition on the Microcontroller:

This implementation ensured edge position, wrist flexion, and wrist extension detections and mapped each gesture to a corresponding rocket movement in the game. It also tracked and reported the number of repetitions performed during the session. The implementation fulfilled the following system requirements: SR0.2, SR2.2, SR3.1, SR3.2, SR4.1, SR4.2, SR9.1.

All signal processing and gesture classification ran entirely on the STM32 MCU embedded in the NeuraLoop device, requiring no external computation.

A control system for identifying edge position, flexion, and extension through HD-sEMG was implemented. A threshold-based strategy

was selected, as threshold-based control has been widely used in myoelectric prosthetic control systems. For example, Billardello et al. (2026)[50] applied amplitude thresholds to distinguish between rest and active muscle states.

To estimate the muscle activation intensity, the acquired 32-channel HD-sEMG signals were preprocessed by full-wave rectification followed by low-pass filtering to obtain an EMG envelope, as proposed by De Luca (1997)[57]. An exponential moving average (EMA) filter was chosen as the low-pass filter [58, 38]. The resulting EMG envelope is given by equation 3, where $x[n]$ is the raw EMG signal, α is the smoothing factor where $0 < \alpha < 1$, and $y[n]$ is the estimated activation envelope. An α of 0.01 was experimentally found.

$$y[n] = (1 - \alpha)y[n - 1] + \alpha|x[n]| \quad (3)$$

With $\alpha = 0.01$ and $f_s = 1000$ Hz, giving a cutoff frequency of:

$$f_c = \frac{f_s}{2\pi} \cos^{-1} \left(1 - \frac{\alpha^2}{2(1 - \alpha)} \right) \approx 1.593 \text{ Hz} \quad (4)$$

The filtered signals were then reduced to a single value per muscle group by computing the arithmetic mean across the three channels with the highest muscle activation for wrist flexion and the three channels with the highest muscle activation for wrist extension, as determined during calibration. Lastly, the difference signal of the flexion and extension envelopes was calculated to be used in the threshold-based control. Instead of applying a single threshold for flexion and extension, a hysteresis threshold-based approach [51] has been chosen to improve robustness against noise EMG fluctuations and prevent rapid oscillations between states. When the difference signal exceeded the positive entry threshold, wrist extension was detected, and the rocket moved rightward, and when it was below the negative entry threshold, wrist flexion was detected and the rocket moved leftward. When the difference signal remained between the exit thresholds without exceeding the entry thresholds, the rest state was detected, and the rocket moved forward.

Throughout the training session, each gesture transition was detected, and the corresponding repetition counter increased for wrist

extension, wrist flexion, and rest, respectively. When the training session ended, the accumulated counts were displayed to remind the patient to document the results (Figure 12).

Latency time:

To facilitate greater agency, the visual feedback must be designed to minimize the delay between physical movement and on-screen response [59]. The latency time was minimized through the implementation of the gesture classification algorithm. Gestures were detected in real-time using threshold-based control, which triggered immediately when the thresholds were crossed. The signal processing was implemented on the MCU. The visual feedback was delivered through rocket movements that were directly mapped to the gesture detection output, ensuring the on-screen response followed the physical movement with minimal delay. The implementation fulfilled the following system requirement: SR0.10.

TEST PROTOCOL

To evaluate the system requirements, the testing process was structured into the following subtests, where each subtest was designed to assess a specific set of system requirements. The NeuraLoop device was attached to the user's right arm before every subtest (Figure 2). The user performing the subtests was a single healthy subject, trained in using the system through the development of the system.

Connection to NeuraLoop and Game Application

This test was designed to evaluate the system's ability to establish a stable connection between the NeuraLoop device and the game application, as well as verifying correct system initialization, user interface feedback, and training session execution. The test further validated following system requirements: SR0.1, SR0.3, SR0.4, SR0.5, SR0.7, SR0.8, SR1.1, SR1.2, and SR7.1

Prior to each test run, a calibration was performed where parameters were provided to the system, consisting of three flexion channels, three extension channels, a baseline threshold, and a hysteresis threshold. The system configuration was kept constant across all trials, and only the training dosage parameter was varied

with 10, 20, 30, 40, 50 meters. The implemented system version supported the **Side to Side** play mode for the right hand.

The test was repeated five times using pre-defined training dosage values of 10, 20, 30, 40, and 50 meters. Each trial started with the NeuraLoop device turned off and disconnected from the computer. The game application was then launched, after which the NeuraLoop device was powered on and connected via the guided connection procedure provided in the application interface. The user entered the calibration parameters into the corresponding input fields before initiating the connection process. Following a successful connection, verified by the status field turning green, the play mode **Side to Side** and **Right Hand** were selected, the training dosage entered, and the session started via the **Play Button** while simultaneously starting an external timer. During the session, gesture visualizations and rocket movement correspondence were verified, along with progression bar advancement. Once the end screen appeared, the timer was stopped. Finally, the NeuraLoop was disconnected and powered off, the status field was verified to turn red, and the application closed.

Microcontroller Processing

This test was designed to evaluate the system's ability to perform signal processing and gesture classification on the MCU of the NeuraLoop device. The test validated the system requirement: SR0.2.

Prior to each test run, a calibration recording was saved and subsequently uploaded in MATLAB. In MATLAB, the differential signal was plotted alongside the gesture classification output, with markers indicating when the entry and exit thresholds were crossed. This allowed visual inspection of whether the gesture labels produced by the MCU classification correspond correctly to the threshold crossings in the differential signal. The test was repeated five times with different calibration recordings.

Calibration

This test was designed to evaluate the system's MATLAB-based calibration tool's ability to calibrate the muscle activity during rest position, wrist flexion and wrist extension. The test validated system requirement: SR0.6.

First, the NeuraLoop device was powered on

and connected to the host computer through WiFi. Subsequently, the calibration was executed, and the participant performed the pre-defined calibration sequence according to the calibration protocol. During the calibration process, the generated calibration interface was inspected to verify that three EMG channels were allocated to wrist flexion and three EMG channels were allocated to wrist extension. The muscle activity of the channels in flexion and extension was visualized in a bar plot, respectively, to ensure that the selected channels represented highest muscle activity. Furthermore, the visualization of the exit threshold and entry threshold was verified by manual selection in order to ensure reliable movement detection. The resulting calibration output was then saved for documentation. After completion of the calibration session, the WiFi connection to the NeuraLoop device was disconnected, the device was powered off, and the calibration window was closed. The test was repeated five times.

Latency Time:

This test was designed to evaluate the time between the user performing flexion, extension, or rest transition, and the corresponding visual feedback through rocket actions in the Unity Game. The test validated system requirement: SR0.13.

The components contributing to the latency time were first signal acquisition, processing, and gesture classification on the NeuraLoop device, then the connection to the computer, and lastly the execution time in Unity. Signal processing was expected to contribute most significantly and was therefore analyzed separately in MATLAB, with a sub-requirement of not exceeding 250 ms to allow sufficient time for the remaining processes.

A calibration recording was used to evaluate signal processing delay. The three most informative channels for flexion and the three most informative channels for extension were selected and processed on the MCU. Since only the EMA envelope filter introduced a delay, the timing difference between the raw rectified EMG signal and the EMA envelope crossing the same threshold was manually inspected in MATLAB. Five flexion transitions and five extension transitions were analyzed by compar-

ing threshold crossing onset times of the raw and EMA filtered signals (Figure 9). The total system latency was evaluated through gameplay video analysis, where gesture onset was defined as the first visible hand movement towards a new position, gesture completion was defined as the moment the final hand position was reached, and system response was defined as the first visible rocket action or stop in the Unity game. To satisfy the SR0.13 requirement, the time from gesture completion to the rocket movement must not exceed 300 ms.

Gesture Recognition

This test was designed to evaluate the system’s ability to detect edge position, wrist flexion and wrist extension, as well as the corresponding in-game rocket movements. It also verifies that the reported number of repetitions matches the detected gestures. The test validated system requirements: SR2.2, SR3.1, SR3.2, SR4.1, SR4.2, SR9.1.

The system supports only edge position, wrist flexion and wrist extension, corresponding to the **Side to Side** play mode in the game. Three states are considered: flexion, extension, and rest, corresponding to edge position. Prior to testing, calibration was performed to achieve the calibration outputs included three flexion channels, three extension channels, a exit threshold, and a hysteresis threshold. The game was opened in Unity and calibration parameters were entered. The system was connected to the NeuraLoop through WiFi, confirmed by a green connection indicator. The **Side to Side** exercise and **Right Hand** were selected, and the training dosage was set to 180 corresponding to 180 seconds before starting the session. During the game, the user started in edge position and changed position corresponding to the arrow highlighted in green and proceeded until the training session was over. Highlighted arrows and detected gestures are logged in Unity during the training session to verify correct detections by comparing the arrow log with the gesture log. A detection was considered successful when each highlighted arrow corresponds to exactly one correctly matched logged gesture. If additional or incorrect gestures were detected during the interval, the transition was classified with the most common false label. After the session, the

total number of repetitions was displayed and compared with logged detections. The NeuraLoop device was then disconnected, powered off, and the game in Unity closed. The test was repeated 2 times.

The results of SR2.2, SR3.1 and SR4.1 was evaluated using a confusion matrix based on comparison of the arrow log with the gesture log in Unity.

Success Criteria and Results:

The success criteria, corresponding system requirements, and test results are summarized in Table 4 and 5.

RESULTS

The results of the tested system requirements in Table 4 and 5 are presented in the following.

Connection to NeuraLoop and Game Application

All implemented requirements for the connection to the NeuraLoop, SR0.3, SR0.4, SR0.5, and all interface requirements for the start screen and game screen; SR0.1, SR0.7, SR0.8, SR1.1, SR1.2, and SR7.1 were successful.

Calibration

The three most informative channels for flexion and extension were highlighted in the calibration (Figure 7). For all five tests, the thresholds were manually adjusted such that the EMG signal activity during the rest gesture fell between the exit thresholds, while the flexion and extension were positioned below and above the entry thresholds, respectively (Figure 8).

Table 2 shows the mean and standard deviation accuracy achieved for rest, flexion, and extension across all five calibration tests.

Gesture	Mean($\pm$ std)%
Rest	95.38($\pm$ 1.78)
Extension	94.92($\pm$ 3.12)
Flexion	95.02($\pm$ 3.03)
Total	95.11($\pm$ 2.52)

Table 2: *Mean classification accuracy and standard deviation of gestures across all five calibration tests.*

Latency time

A total of 21 gesture transitions were analyzed from a gameplay video. The mean ges-

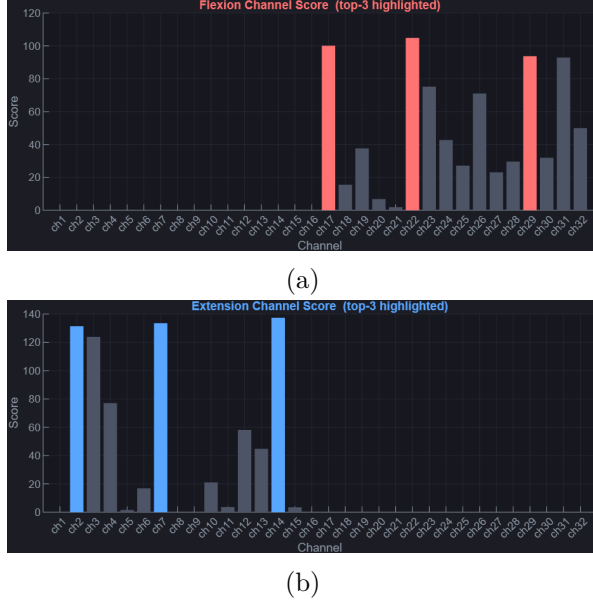


Figure 7: Calibration interface showing channel selection from a test iteration. Red channels indicate the three most informative channels for flexion (a), and blue channels indicate the three most informative channels for extension (b). The x-axis shows EMG channels and the y-axis shows the corresponding contrast score.

ture duration was $314.69(\pm 108.45)ms$. The mean system reaction time measured from gesture onset was $308.33(\pm 193.44)ms$, while the mean reaction time measured from the gesture completion was $-6.36(\pm 194.51)ms$. Regarding signal processing delay, the average delay introduced by the EMA envelope filter was $132.3(\pm 57.0)ms$ (Figure 9).

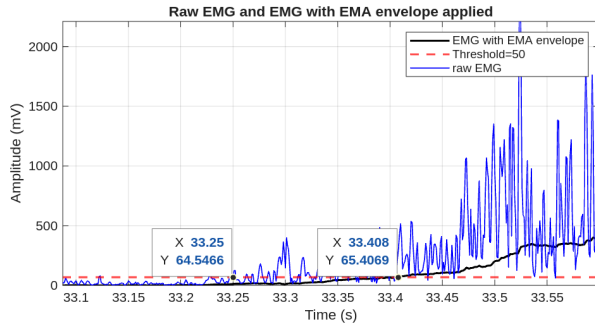


Figure 9: Delay introduced by Exponential Moving Average envelope filter with $\alpha = 0.01$

Microcontroller Processing

The tests showed that the gesture classification output from the MCU was aligned with

the corresponding difference signal, also calculated on the MCU. The expected state transitions, depending on the difference signal and the actual state transition of the gesture classification, was aligned with an accuracy of 100% within 100 ms for all five tests. Figure 10 shows the difference signal and the gesture classification, where the gesture classification of flexion was assigned the value -100, extension 100, and rest 0.

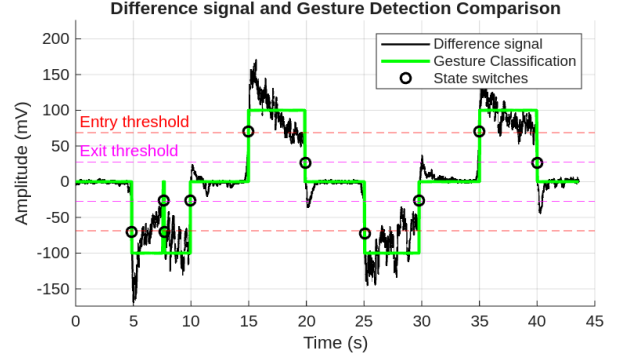


Figure 10: Difference EMG signal and the gesture classification on the microcontroller with marked the state switches.

Gesture classification performance

A total of 176 gesture transitions was analyzed and recorded across two recording sessions of three minutes each. The gesture sequence alternated between rest, flexion, and extension, where the initial rest period was excluded from the analysis. Consequently, expected gesture transitions per session, and gesture transitions for the first session and second session are shown in Table 3.

	Expected	Session 1	Session 2
Flexion	22	41	30
Extension	22	39	23
Rest	44	78	52
Total	88	158	105

Table 3: Expected gesture transitions and actual gesture transitions detected in sessions 1 and 2.

The global system had an accuracy of 72.73%, a precision of 72.66%, and a recall of 72.73%. The recall for flexion, extension, and edge was respectively 53%, 74%, and 91% (Figure 11).

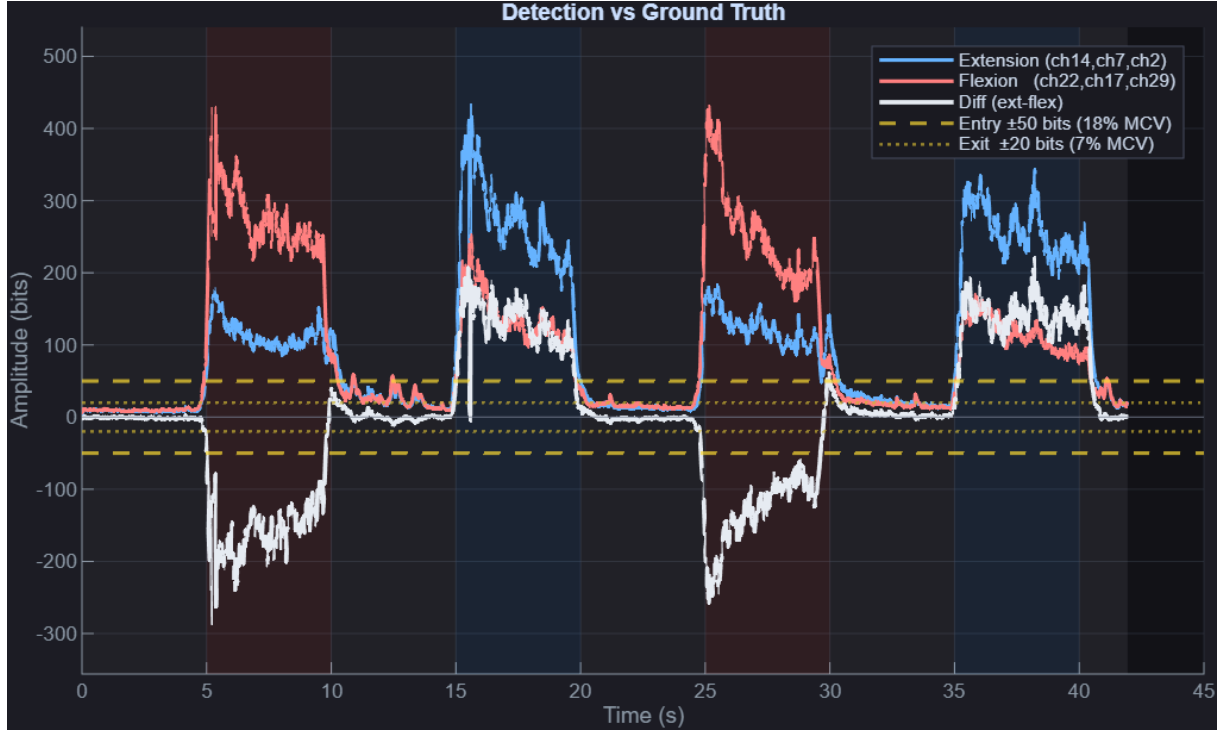


Figure 8: Calibration interface figure from test iteration number 3, where the entry threshold was set to 50 bits, and the exit threshold to 20 bits.

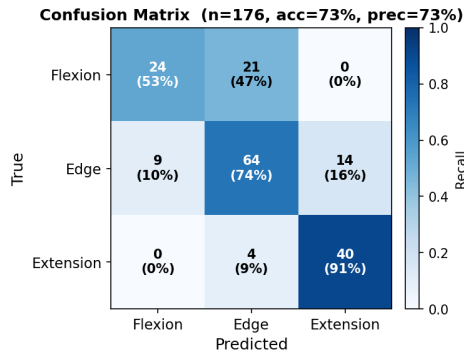
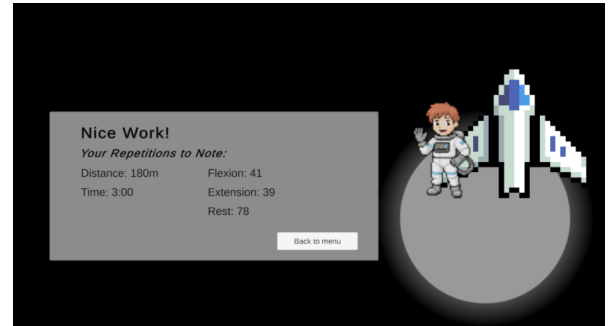


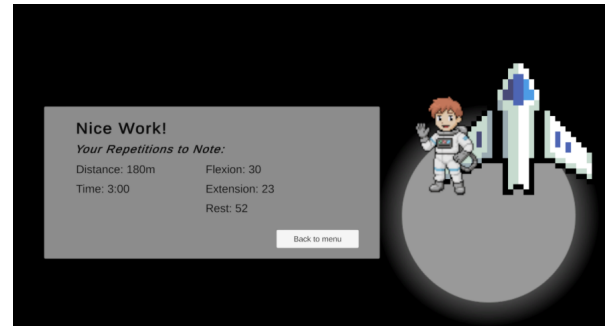
Figure 11: Confusion matrix of the gesture detection during the 176 gesture transitions. The recall for each hand gesture is presented diagonally, and the number of gestures(n), accuracy(acc), and precision(prec) are presented above the matrix

During the two 3-minute recordings, the rocket's movement was manually observed, where the rocket moved to the left when wrist flexion was detected and to the right when wrist extension was detected.

The detected gestures for session 1 and session 2 in Table 3 agree with the displayed detected gestures of the end screen (Figure 12).



(a)



(b)

Figure 12: The end screen displaying the detected gestures in test session 1 (a) and 2 (b).

Table 4: *Implemented system requirements not derived from the user requirements prioritized in according to MoSCoW. The IDs of these requirements were assigned starting with 0.*

ID	Requirement	MoSCoW	Success Criteria	Result	Passed
SR0.1	The system must be an application that can be started by clicking or tapping it.	Must	The application opens successfully.	5/5	Pass
SR0.2	The system must perform signal processing and gesture classification on the NeuraLoop device microcontroller.	Must	The gesture byte changes when the signal exceeds or falls below the entry threshold successfully.	5/5	Pass
SR0.3	The system must show a guide to connect with NeuraLoop.	Must	The system shows a guide to connect with NeuraLoop, and the connection can be established by following the guide.	5/5	Pass
SR0.4	The system must be able to connect with NeuraLoop.	Must	The system connect to the NeuraLoop.	5/5	Pass
SR0.5	The system must show whether the connection to the NeuraLoop is established.	Must	The system shows whether the connection to the NeuraLoop is established.	5/5	Pass
SR0.6	The system must be able to adapt to the muscle activity level of the user through a calibration before starting the training session.	Must	The calibration results correctly visualized three flexion and three extension channels in a plot together with the corresponding thresholds. The classification accuracies for the edge position, wrist flexion, and wrist extension > 70%.	5/5	Pass
SR0.7	The system must contain a function to start the training session.	Must	The Play Button start the training session when activated.	5/5	Pass
SR0.8	The system must track and show the completed training dosage and calculate the remain training dosage during training session.	Should	The system tracked and showed the completed training dosage and calculated the remaining training dosage during the training session by filling the progress bar continuously.	5/5	Pass
SR0.10	The system's latency time from performing the movement to give visual feedback on the movement must be kept below 300 ms.	Should	Time from gesture completion to rocket movement must not exceed 300ms, evaluated over 5 flexion and 5 extension transitions.	10/10	Pass

Table 5: *Implemented system requirements derived from the user requirements prioritized according to MoSCoW. System requirements derived from a specific user requirement were assigned IDs beginning with the corresponding user requirement number.*

ID	Requirement	MoSCoW	Success Criteria	Result	Passed
SR1.1	The system must provide an interface where the therapist can select the exercise combination: EFE, EFEC, EFEO, EFECO.	Must	The system provided an interface to select the exercise combination, and the EFE exercise combination could be selected.	5/5	Pass
SR1.2	The system must provide an interface where the therapist can select the training dosage, and the system must adjust the training session according to the selected dosage and end the session once the selected dosage is completed.	Should	The system provided an interface where the training dosage could be entered, adjusted the training session according to the selected dosage, and ended the session within 3 seconds once the selected dosage was completed.	5/5	Pass
SR2.2	The system must be able to detect the edge position.	Must	Classification accuracy > 70%. A detection is successful if when the interval the up arrow appears has only one logged edge position.	74%	Pass
SR3.1	The system must be able to detect wrist flexion when wrist flexion is performed or attempted to be performed.	Must	Classification accuracy > 70% A detection is successful if when the interval the left arrow appears has only one logged wrist flexion.	53%	Fail
SR3.2	In the movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the left when wrist flexion is detected.	Must	Manual verification of the rocket moving to the left when wrist flexion is detected during a 180-second training session.	2/2	Pass
SR4.1	The system must be able to detect wrist extension when wrist extension is performed or attempted to be performed.	Must	Classification accuracy > 70%. A detection is successful if when the interval the right arrow appears has only one logged wrist extension.	91%	Pass
SR4.2	In the movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the right when wrist extension is detected.	Must	Manual verification of the rocket moving to the right when wrist extension is detected during a 180-second training session.	2/2	Pass
SR7.1	In the movement combinations EFE, EFEC, EFEO, EFECO, the system must show a visualization of the hand movements wrist flexion, wrist extension, edge position, closing hand, and opening hand paired with the corresponding rocket actions.	Must	The system showed a visualization of the hand movements wrist flexion, wrist extension, and edge position paired with their corresponding rocket actions.	5/5	Pass
SR9.1	The system must inform the user of the achieved number of repetitions corresponding to the detected gestures when the training session is ended.	Could	The system showed the achieved number of repetitions on the end screen matching the logged gestures.	2/2	Pass

DISCUSSION

This work aimed to develop an MCU-based EMG-controlled gamification system using the NeuraLoop for the rehabilitation of hand function after stroke.

Overall, the results demonstrated that embedded EMG-based gesture recognition can be used to provide real-time control of a rehabilitation game. The developed system combined wearable HD-sEMG acquisition, embedded gesture classification, and a game-based training environment to create a less therapist-supervised rehabilitation tool, with the purpose of being used in a rehabilitation training setting. A game was developed in which the user controls a rocket by moving their hand, navigating left and right to avoid incoming asteroids. Hand movements were detected in real-time through EMG signals by using a threshold-based gesture classification algorithm implemented on the MCU in the NeuraLoop device. Training dosage was monitored through a game progress bar, and the accumulated movement counts were presented at the end of each training session to support the user in documenting the achieved training repetitions.

Murciego et al.(2025) highlighted that embedded signal processing and gesture recognition would improve the usability and portability of rehabilitation systems. This work implemented threshold-based gesture control on the NeuraLoop device’s MCU, enabling fully embedded, real-time control of the hand rehabilitation game. The embedded gesture state transitions, based on the decision input derived from the muscle activity in the flexor and extensor muscles, were fully aligned with the computer-simulated state transitions within 100 ms (Figure 10). This indicated that the gesture control on the MCU was reliable according to the input decision signal.

To address earlier reported barriers as challenges with electrode placement, setup difficulties, and system complexity, reported by Munoz-Novoa et al. (2024). Multiple design choices have been made. The choice of using the NeuraLoop technology, including the HD-sEMG sleeve, enabled a grid of measure-points around the forearm, reducing the risk of miss-

ing important muscle activity areas, thereby requiring less precision from the user to set up. Furthermore, the calibration using the contrast score-based channel selection prioritizes informative channels with high contrast between flexion and extension. This provides robustness against suboptimal sleeve placement, as the algorithm is less sensitive to skewed positioning, particularly relevant for users with smaller arms, where the risk of electrode overlap is greater. The choice of the HD-sEMG sleeve and the calibration are particularly advantageous to solve the barriers of setup difficulties emphasized by Munoz-Novoa et al. (2024), making the rehabilitation system less therapist-supervised, self-contained, and likely more suitable for practical deployment.

However, the calibration relies on MVC for establishing reference EMG levels. Stronger MVC performance produces more distinct channel activation, enabling clearer thresholds. In practice, MVC levels are not expected to be consistently reached during the training session due to the increased fatigue commonly observed in stroke patients [60]. If thresholds are set too close to MVC levels, gestures may go undetected or misclassified when activation falls below them. Thresholds must therefore reflect expected activation levels during the training session.

While these design choices improved usability and robustness of the system setup, the practical value of the platform depends on how reliably the embedded classifier can detect intended movements during gameplay. Therefore, the gesture classification performance was evaluated. The overall accuracy of gesture classification during the game was 73% with accuracy of flexion being 53% and extension 91%. The considerable difference between flexion and extension indicates that the flexion signal is more susceptible to fluctuations than the extension (Figure 10), leading to a misclassification of 47% as rest (Figure 11). The lower accuracy of flexion could suggest an asymmetry in the discriminative power of the recorded muscle activity, even though this is not clear in Figure 8, the difference signal in Figure 10 indicates that the negative parts, the parts where the user performed flexion, fluctuate more around the threshold placements. Consequently, almost

half of all intended flexion gestures were classified as rest, which became the primary factor limiting the overall classification performance with an accuracy of 73%.

As highlighted by Roche et al.(2014), threshold-based methods remain widely preferred in real-world prosthetic and rehabilitation systems due to their reliability, minimal training requirements, and computational efficiency. However, direct comparison with threshold-based methods is challenging as the identified studies utilizing threshold-based gesture classification [38, 47, 48, 49, 50], do not report standardized performance metrics such as accuracy or recall, making it difficult to benchmark against methodologically similar approaches. Therefore, the following comparison focuses on machine-learning-based gesture classification systems, while acknowledging the fundamental methodological differences.

Although the achieved performance demonstrates the feasibility of embedded threshold-based gesture recognition, the obtained accuracy remains lower than that reported in studies employing machine learning approaches. Murciego et al. (2022, 2023) achieved accuracy above 90% using LDA for hand gesture classification, and Xu et al. (2016) reported 95% accuracy with LDA-based EMG classification implemented on a MCU in real-time. Benatti et al. (2015) similarly achieved 94% using an SVM-based solution on a MCU. Regarding recall, the implemented system achieved 73%, whereas Murciego et al.(2025) reported a recall of 97.1% using an SVM classifier for micro-gesture detection. The performance gap is likely related to the fundamental differences between threshold-based and machine-learning approaches. Whereas the threshold-based method relies on predefined decision boundaries [51], machine learning classifiers can model more complex signal characteristics [45]. However, these gains in accuracy come at the cost of increased implementation complexity, training requirements, and computational demands. The choice of a threshold-based approach in this work was therefore motivated by the desire for transparency, low computational cost, and straightforward deployment on the embedded hardware.

An important limitation when interpreting

the obtained classification performance is that the system was evaluated on a single healthy subject rather than multiple stroke patients. Since the intended target population has expected altered activation patterns, including increased flexor activity, co-activation of flexor and extensor muscle groups, and impaired selective motor control [17, 18, 19], gesture discrimination may be more challenging. In stroke patients, it is expected that the difference signal used for gesture classification would likely exhibit a negative baseline offset due to persistent flexor activity, and not centered around zero as for the healthy subject (Figure 10). Furthermore, co-activation of flexor and extensor muscles may reduce the separation between movement classes, resulting in smaller deviation from baseline during intentional flexion and extension. Consequently, the classification performance reported in this work should be interpreted as a proof-of-concept demonstration of the technical feasibility of embedded gesture recognition for a rehabilitation game rather than an estimate of expected performance in the target population.

Beyond classification accuracy, the responsiveness of the system is critical for maintaining a natural interaction between the user and the rehabilitation system[61]. Therefore, the temporal performance of the complete processing pipeline was evaluated.

The mean system reaction time from gesture completion was $-6.36(\pm 194.51)ms$, indicating that the system responded approximately 6 ms prior to gesture completion, suggesting that the classifier was able to predict the intended gesture before its natural endpoint. This implied that the system does not introduce a perceptible delay relative to the user’s movement. From a rehabilitation perspective, this level of responsiveness is important because delayed feedback can reduce the sense of agency and negatively affect user engagement [59]. The observed reaction time supports the suitability of the system for interactive game-based training.

In the qualitative interview, the importance of thumb activation when opening and closing the hand was emphasized, but the implemented system consists of only flexion and extension training. Therefore, in further development

of the game, it would be relevant to integrate hand movement with a focus on thumb activation and opening and closing the hand, leading to the selection of several exercise combinations. This would support training in reaching and grasping objects in ADL [17, 18], and the possibility of adjusting the training to the individual user’s ability level. Additionally, the current system is implemented only for right-hand users. However, it would also be relevant to develop a left-hand version of the game to support a wider range of stroke patients. This is important as stroke affecting the right hemisphere also results in left-sided hemiparesis, ataxia, and impaired movement coordination [62].

By expanding the game actions to include the entire system (Figure 3), it is expected to increase the engagement of the game, which is essential for effective EMG-based rehabilitation, as Mejía et al. (2019) highlight that intuitive game design is key to maintaining engagement and usability.

Future Perspective

In future versions, it would be beneficial to investigate whether LDA or SVM machine learning approaches are suitable for the MCU application. To evaluate how the pathological muscle activation patterns in stroke patients differ from those of healthy subjects and affect signal characteristics, threshold adaptation, and overall gesture classification performance. Future work should include testing on multiple stroke patients. To decrease the system complexity and increase the usability, the calibration must be implemented in the game application. This will reduce the complexity by reducing the number of programs the user has to interact with to only the game application, and thereby increase the usability of the system. Further development of the game, should investigate how to integrate hand movement with a focus on thumb activation and opening and closing the hand, to further support training in reaching and grasping objects in ADL. To increase the engagement of the system, expanding the game actions and environment is suggested.

CONCLUSION

This work developed an MCU-based EMG-controlled gamification system utilizing the NeuraLoop technology for the rehabilitation of hand function following stroke. A threshold-based gesture classification algorithm was implemented on the MCU to detect hand movements in real-time using the envelope of EMG-measured muscle activity. The classification outputs were combined into a rehabilitation game where the user controlled a rocket in a space environment. The signal processing and gesture classification embedded on the MCU showed full agreement with the expected state transitions. Overall gesture classification accuracy during the game reached 73%. The rocket actions mapped to the classification responded without errors when wrist flexion, wrist extension, and edge position were detected. These results demonstrated the feasibility of an MCU-based EMG-controlled system for interactive rehabilitation gaming. Future work should investigate methods to improve gesture classification accuracy, such as LDA or SVM.

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A Problem Analysis

A.1 Acute Stroke: Epidemiology, Diagnosis and Treatment

Cerebrovascular accident (CVA) or stroke is defined as a neurological disorder caused by impaired cerebral blood perfusion [1, 2]. According to the International Classification of Diseases Eleventh Revision (ICD11), stroke is categorized into intracerebral haemorrhage, subarachnoid haemorrhage, cerebral ischaemic stroke, and stroke of undetermined type [63]. Ischemic stroke is caused by a blockage of a blood vessel, and haemorrhage stroke is caused by bleeding within or around the brain tissue [1, 2]. Both conditions can lead to the sudden death of brain cells due to insufficient oxygen delivery, increased intracranial pressure, inflammation, and edema [1, 2].

Stroke commonly presents with sudden focal neurological deficits, including unilateral weakness of the arm or leg, speech disturbance, confusion, impaired balance, dizziness, and vision changes [64, 65, 66]. Early recognition of these symptoms is essential for rapid clinical assessment and diagnosis, as timely neuroimaging and treatment initiation are critical in acute stroke management.[64, 67]. The most recent Global Burden of Disease Study 2021 [3] reported 11.9 million incident stroke cases worldwide, with ischemic stroke as the most common subtype, accounting for approximately 65%. They estimate that 93.8 million individuals were living with stroke-related disability, making stroke the fourth most common cause of disability-adjusted life years globally. The risk of stroke increases with age [3], and doubles over the age of 55 for both women and men [1]. In addition, several risk factors contribute to stroke development, including hypertension, diabetes, smoking, alcohol, drug abuse, physical inactivity, and poor diet [1]. Although stroke incidence is generally highest among older adults, increasing incidence has been observed in the age range between 20-54 years [1].

Rapid diagnosis and reperfusion are essential in the management of acute ischemic stroke to minimize brain injury. The 2026 AHA/ASA guideline emphasizes that neuroimaging is central in the early evaluation of patients with suspected acute ischemic stroke [67]. Initial brain imaging aims to distinguish ischemic stroke from intracranial haemorrhage, assess infarct extent, detect large vessel occlusion (LVO), and determine eligibility for reperfusion therapy[67]. Non-Contrast Computed Tomography (NCCT) is recommended as the first-line imaging modality in the acute setting, due to its rapid and high sensitivity for detecting intracranial haemorrhage[67]. Exclusion of haemorrhage is essential before intravenous thrombolysis (IVT). Furthermore, NCCT can identify early ischemic changes and is generally sufficient for guiding IVT decisions within 4.5 hours from symptom onset [67, 68]. In acute ischemic stroke, reperfusion therapy primarily consists of IVT, which facilitates clot dissolution and restoration of cerebral blood flow [1]. The effectiveness of IVT is strongly time-dependent, emphasizing the importance of rapid treatment initiation [1, 68]. In patients with LVO, mechanical thrombectomy may additionally be performed to physically remove the occluding thrombus using endovascular techniques. [67, 69]. Most patients with stroke survive the initial illness, and the long-term consequences often cause the greatest health burden [4].

A.2 Consequences of Stroke: Cognitive, Motor, and Functional Impairments

Approximately 35-40% of stroke survivors experience limitations in activities of daily living (ADL) six months after the stroke event [5]. In addition to physical impairments, stroke frequently results in cognitive deficits. A review by Cramer et al.(2023) identifies attention deficits, loss of executive function, memory disorders, neglect, aphasia, and apraxia as common cognitive syndromes that affect functional independence and task performance. Deficits in attention and executive functioning may impair concentration, information processing, planning, decision-making, and the ability to perform sequential or multitasking activities [10]. Individuals may also experience reduced processing speed, mental fatigue, impaired spatial awareness, and difficulties

with comprehension and communication [10, 62]. Aphasia can affect both speech production and language understanding, while apraxia impairs the ability to execute purposeful movements despite preserved motor capacity [10, 62]. Furthermore, stroke affecting the left cerebral hemisphere are commonly associated with aphasia, difficulties in speech production and language comprehension, whereas stroke affecting the right cerebral hemisphere may result in impaired spatial awareness, difficulties in coordinating movement [62].

One of the most common impairments following stroke is motor impairment, which affects muscle function and mobility [4]. Stroke affects lower extremity physical function, including balance, gait function, and gait speed, which are crucial for independence in daily activities and quality of life [70]. Balance impairment affects approximately 48% of stroke survivors, consistently identified as a significant risk factor for falls and decreased independence in daily activities [71]. Together with cognitive deficits, these impairments collectively limit functional independence and quality of life. Among the most functionally disabling consequences is upper limb paresis, affecting more than 70% of stroke survivors and severely limiting the ability to perform daily activities [72]. Given its high prevalence and profound impact on independence, upper limb motor impairment is the primary focus in this work.

A.3 Hemiparesis and Upper Limb Impairments After Stroke

Motor impairment is often caused by hemiparesis, which presents as muscle weakness on the contralateral side of the body relative to the cerebral lesion. Hemiparesis is defined as reduced ability to generate a normal level of muscle strength [6, 62]. Paresis results in abnormal posture and stretch reflexes, and impaired voluntary movement [6]. The functional consequences of stroke depend on the affected brain region, as different areas of the brain are responsible for specific functions [73].

Hemiparesis affecting the upper extremity often results in significant upper limb motor impairment [74]. Upper limb motor impairment affects more than 55% of chronic stroke survivors, and at least 70% experience functional limitations that reduce independence in ADL [26, 75, 76]. Such motor dysfunction contributes to reduced ADL performance, reduced mobility, impaired social participation, and lower quality of life [77, 78].

The performance of ADL depends on the coordinated interaction between hand and arm function. For example, during drinking tasks, the hand is responsible for grasping the object, while the arm enables reaching and stabilizing movement throughout the activity [8, 9]. Upper limb motor recovery following stroke generally follows a proximal-to-distal pattern, with hand function often showing the greatest impairment [9].

For improvement of the upper limb function and independence in ADL, rehabilitation programs commonly include intensive and repetitive task-specific training exercises for stroke patients [79]. Several studies highlight that high doses of repetitive task-specific practice can significantly improve upper limb function in chronic stroke patients [80, 81, 82].

A.4 Upper Limb Motor Rehabilitation post Stroke

As described in section A.3, stroke frequently results in upper limb impairment including reduced motor control, sensory deficits, and proprioceptive loss, which can lead to contracture, paresis, and spasticity [7]. Spasticity is a motor control disorder and a frequent consequence of stroke, affecting 20-40% of patients and is a major contributor to morbidity and disability [83], leading to functional impairments and restrictions in ADL [18]. Pharmacological management of spasticity includes Botulinum Toxin A (BTX-A) injections, which is especially effective in reducing spasticity in elbow pronators, wrist and finger flexors [83]. Oral treatment against spasticity such as baclofen and tizanidine are also used which improve the motor function and reduces muscle tone [18].

Upper limb rehabilitation is delivered by physical therapists and occupational therapists at the hospitalization and in the early supported training at home, with some patients transferred to rehabilitation centres once medically stable [7]. The rehabilitation process includes assessing the patient's needs, setting measurable goals for improvement, implementing targeted interventions, and reassessing progress [4]. Treatment is delivered by a coordinated multidisciplinary team, emphasizing high intensity, task-specific training through repeated practice of functional activities to improve functionality and independence [4].

The Fugl-Meyer Assessment of the Upper Extremity (FMA-UE) is the most widely used measure of upper extremity motor impairment after stroke, assessing coordination and reflex activity across the shoulder, elbow, forearm, wrist and hand, and classifying impairment severity as hemiplegia, hemiparesis, or slight motor dyscoordination [84, 85]. Additionally, the FMA-UE is the gold-standard for planning rehabilitation practice by evaluating performance of 33 items, where each item was assessed in a scale range from 0-2 [86]. The severity is defined based on the score of the 33 items, where severe < 30 , moderate $30 \leq 45$, and mild > 45 [87]. The assessment is used before and after treatment to evaluate changes in upper extremity motor function [87].

A.4.1 Occupational Therapy of Upper Limb Motor Impairments

The rehabilitation after stroke is beginning when the medical condition is stable [11]. Occupational therapy is central to facilitate post stroke patient participation in ADL [5, 11]. Rowland et al. (2008) describe that occupational therapy including individually selected and graded tasks and activities. For the rehabilitation process, the occupational therapists assess motor and sensory changes after stroke, focusing especially on upper limb and hand function. The initial process involving interviews with the patient to get insight in the ADL by observing the limitations of the activities.

Interventions improving performance in ADL most effective is including mirror therapy, motor imagery and task-oriented training [5]. Mirror therapy involves performing movements with the unaffected arm while a mirror is placed in the sagittal plane, creating the visual illusion that the affected arm is moving normally [88]. This visual feedback is intended to stimulate the brain and promote neural reorganization after stroke [88]. Saragih et al. (2025) demonstrated that mirror therapy improved the hand function of stroke patients based on the FMA-UE [89]. Motor imagery involves mentally rehearsing a movement without physically performing it [90]. It relies on cognitive and sensory processes that allow the individual to activate representations in working memory [90]. This mental rehearsal helps improve motor performance and functional abilities [90]. Task-oriented training focuses on practicing meaningful, goal-directed everyday activities, such as reaching for a cup or buttoning a shirt [91]. By training whole tasks in realistic contexts, task-oriented training aims to improve the transfer of skills to daily life [91]. Evidence suggests that the most effective interventions for upper-limb recovery after stroke are based on repetitive, task-specific motor learning principles [12]. One rehabilitation approach that incorporates these principles is constraint-induced movement therapy (CIMT). In mildly impaired stroke survivors, approximately 20–25%, CIMT has been shown to achieve greater improvements in hand and arm function than conventional therapy [15]. CIMT and its modified version (mCIMT) are among the most effective physical therapy approaches for improving function in the paretic upper limb [13]. CIMT is a rehabilitation approach that involves restraining the non-paretic arm for most waking hours, providing around six hours of intensive and repetitive task training daily in 2 weeks, and using behavioral strategies to facilitate adherence and transfer of gains to ADL [13, 12]. The modified version, mCIMT, is defined as a less intensive version of CIMT, where the daily training dosage is reduced, typically ranging from 30 minutes to 6 hours per day over 2–12 weeks [12, 13]. Kwakkel et al. (2015) reported strong evidence that both CIMT and mCIMT improve motor function, arm-hand activities, and functional use of the affected upper limb following stroke, with effects sustained over time.

A.4.2 Recent Technologies for Upper Limb Motor Rehabilitation

For upper limb rehabilitation following stroke, assistive technologies are widely used to enable high-intensity repetitive training [92]. A selection of technologies is presented to illustrate different approaches within robot-assisted and technology supported, and the following section presents selected examples to illustrate their clinical application and functional focus.

Raciti et al. (2025) [93] presented AMADEO robotic system, a rehabilitation system designed for hand and finger function recovery in neurological conditions as stroke where each finger is attached individually to a robotic actuator. The AMADEO system is integrated with interactive Virtual Reality (VR) games where real-time grip strength, range of motion, and coordination is monitored. The AMADEO system aims to promote neuroplasticity and upper limb recovery following stroke by delivering high-dose, repetitive, and goal-directed fine motor training. In contrast to hand training with AMADEO, the Armeo Spring targets the proximal upper limb function. It is a passive, gravity-supporting exoskeletal device, that enables patients to perform active reaching and grasping exercises within a virtual reality environment. Colomer et al. (2013) [94] reported significant improvements in motor function and daily activity in chronic stroke patients with mild to moderate hemiparesis following 36 sessions with the Armeo Spring, with effects largely maintained four months after treatment completion.

Beyond robotic exoskeleton systems, wearable technologies have also been introduced. Imtiaz et al. (2023) [95] developed a smart glove for post-stroke rehabilitation. The glove uses sensors to detect hand movements and provide user feedback, while a micro-controller processes the data to help restore natural movements. By connecting the glove to a computer or mobile device, individually exercises are customized to improve hand mobility, strength, and dexterity. The smart glove is non-invasive, user-friendly, and cost-effective, making it suitable for stroke rehabilitation. Similarly, soft robotic gloves (SRG) have been investigated as an emerging rehabilitation tool. In a systematic review of Ko et al. (2023) [96] compared SRG-based interventions with conventional rehabilitation approaches. The SRG interventions varied across studies, incorporating biofeedback, task-oriented movement training, and in some cases game-based environments. The results of the systematic review indicate that using SRG in rehabilitation improved the FMA-UE scores. Additionally, patients demonstrate better immediate effects following SRG intervention.

These studies demonstrated potential of technologies to assist in upper limb rehabilitation. The systems are generally defined as high-cost technologies which limits the accessibility in many clinical and home settings. Furthermore, several technologies require continuous supervision by healthcare professionals which excluding semi-supervised or independent training.

A.5 EMG-Controlled Gamification Systems in Upper Limb Motor Rehabilitation

EMG-controlled serious games and virtual reality (VR) systems have demonstrated high levels of engagement and motivation for stroke rehabilitation [26, 25, 27, 28]. In these systems surface EMG signals from forearm muscles are used as a human-machine interface, where muscle activation is translated into control commands for computer games or virtual environments. This enables stroke patients to perform task-specific movements while receiving immediate visual feedback, supporting motor learning and active participation in rehabilitation exercises [25, 26, 29, 30]. Ito et al. (2023, 2024) further reported increased engagement with repetitive tasks compared to conventional exercises.

For upper limb stroke rehabilitation, studies has highlighted gamified EMG-Controlled systems to improve clinical outcomes but also user experience, motivation and adherence to training program. An overview of the studies is presented in Table 15.

Several studies report high levels of patient engagement and motivation using EMG controlled rehabilitation games [31, 32, 33, 34, 35]. Ito et al. (2023) [25] observed that stroke patients rated EMG-controlled games as more engaging than conventional exercises and expressed willingness

for continued use. Additionally, Ito et al. (2024) [26] further emphasized that gamification with adaptive difficulty helped maintain engagement during otherwise repetitive training where participants reporting high enjoyment.

Similarly, Dash and Lahiri (2020) [97] reported high acceptance of VR feedback-based rehabilitation among stroke patients, where clinical outcomes has shown improved grip strength and task performance. Farasyfa et al. (2022)[27] demonstrated that VR enhanced motivation during wrist movement training controlled by EMG signals. These findings suggest that game-based interfaces meaningfully reduce the perceived burden of repetitive training tasks by improving the motivation and adherence.

Studies by Marin-Pardo et al. (2022, 2023, 2024) [31, 32, 33] presented Tele-REINVENT, demonstrating that VR based serious games combined with real-time sEMG feedback lead to high adherence and motivation among chronic stroke patients where participants completed a 6-weeks rehabilitation program including 30 training session. A feasibility study by Marin-Pardo et al. (2022) showed improvements in neuromuscular activation and functional performance following the 6-week training program, while a subsequent qualitative analysis by Marin-Pardo et al. (2023) reported high user acceptability and adherence. More recent findings by Marin-Pardo et al. (2024) further indicates that gains in motor control are associated with improvements in quality of life among chronic stroke patients. Donoso Brown et al. (2014, 2015) [29, 28] similarly demonstrated a 4-week home-based EMG biofeedback training program using a commercial game controlled by wrist flexor and extensor EMG signals. The study by Donosa Brown et al. (2014) achieved an increased muscle activation and voluntary control during EMG based gamification training program. Interview from the study by Donosa Brown et al. (2015) further reinforces that the EMG based training program increased engagement, awareness and use of the affected limb in ADL.

Beside the positive user experience, studies has identified technical challenges when using the system. Munoz-Novoa et al. (2024) [98] presented practical barriers including electrode placement difficulties, system complexity and need for assistance during home use. These challenges may affect patients with motor impairment or limited technological literacy. In according to this, Mejía et al.(2019) [36] emphasized that intuitive and simple game design is essential for effective EMG-based rehabilitation, as complex interfaces and practical barriers risk reduced engagement and usability.

Related to clinical outcomes, EMG-controlled gamified interventions have demonstrated measurable improvements of functionality. A randomized controlled trial by Amin et al. (2025) [34] reinforces these findings by showing that patients receiving EMG-based VR training with gamified elements achieved significantly greater improvements in hand function, dexterity, grip strength, and range of motion compared to a control group receiving conventional therapy. This indicates that gamification not only influences motivation but also enhance rehabilitation outcomes. Seo et al. (2022) [35] reported that, following nine sessions of EMG game training, stroke survivors showed a 46% reduction in the time needed to move a cursor to a novel sequence of four target locations on a 5×5 grid, a test performed post-training using the paretic hand, and a 45% reduction in normalized path length (actual cursor path relative to the minimum straight-line distance between targets), indicating more efficient and direct control of muscle activation patterns. Ghassemi et al. (2019) [99] demonstrated a 67% improvement in movement efficiency among healthy participants using EMG-controlled cursor games, suggesting strong potential for translating such approaches to clinical outcomes.

Song et al. (2022) [100, 101] investigated a wearable multi-sensor system integrating EMG, force myography (FMG), and an inertial measurement unit (IMU) to classify 12 ADL-related hand movements in stroke patients for use as serious game input. Combining all three modalities outperformed any single sensing modality alone, and patients reported greater enthusiasm and motivation for rehabilitation compared to conventional exercise.

These findings indicate that EMG-based gamified rehabilitation can simultaneously support

clinical recovery and enhance the subjective rehabilitation experience.

A.6 Hand Movement and Alterations Post Stroke

A.6.1 Forearm Muscle Construction Contributing to Hand Function

To perform ADL and essential activities in work life, grasping and hand manipulations are fundamental [102]. These motor abilities require coordinated activation of the approximately 38 muscles in the forearm, wrist, hand, and fingers, allowing complex interactions and movements of the joints, which gives us 25 degrees of freedom [102, 103].

The muscles in the forearm and hand are divided into extrinsic and intrinsic muscle groups, see figure 13. The extrinsic muscles originate in the forearm and insert into the hand and fingers, whereas the intrinsic muscles are located entirely in the hand [102, 104]. The main purpose of the extrinsic muscles is to generate large forces and gross movements necessary for grasping, object manipulation, wrist stabilization, supination and pronation of the forearm [102]. In contrast, the intrinsic muscles are responsible for fine motor control, precision grips, and the coordination of finger movements [102]. Since many of the most basic hand functions are heavily dependent on the extrinsic muscles [102], the following sections will focus on the extrinsic muscle group of the hand.

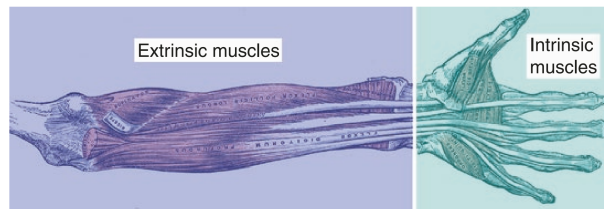


Figure 13: Division of the forearm muscles into the extrinsic and intrinsic muscle groups [104]

A.6.2 Extrinsic Muscles and Their Functional Roles

The forearm extends from the elbow to the wrist and contains two bones, the radius and ulna [16, 105]. To keep the hand and fingers small, compact, and precise, most of the powerful muscles controlling the wrist and fingers are located in the forearm rather than in the hand itself [16]. These muscles are referred to as extrinsic muscles and consist of 19 muscles [105, 106]. Many of the extrinsic muscles originate from the humerus, the long bone in the upper arm, allowing the extrinsic muscles to have longer muscle fibers and tendons, and to provide larger attachment areas [16].

The extrinsic muscles are generally divided into flexor and extensor muscles[16]. The flexor muscles are mainly placed in the anterior compartment and are connected to the anterior surface of the wrist and hand [16]. In contrast, the extensor muscles are mainly located in the posterior compartment, attach to the posterior side of the wrist and hand [16]. Generally, the flexor muscles are stronger than the extensor muscles and are the most important for grasping and force generation [16].

Both the flexor and the extensor muscles are anatomically organized in layers, where the flexor muscles are organized in three layers, superficial, intermediate, and deep layer muscles, and the extensor muscles are organized into two layers, superficial and deep layer muscles [16, 105]. The superficial layer is the muscles nearest the skin, the deep muscles are the muscles under the superficial layer nearest the bones, and the intermediate layer is between the superficial and the deep layer [16, 105].

The flexor muscles can further be grouped according to their primary function, including forearm, pronation, wrist flexion, finger flexion, and thumb flexion, as shown in Table 6. Although the brachioradialis is anatomically located in the posterior compartment of the forearm,

Muscle	Primary Function	Layer
<i>1. Muscles that pronate the forearm and flex the elbow</i>		
Pronator teres	Forearm pronation and assists elbow flexion.	Superficial
Brachioradialis	Flexes the elbow in edge position, see Figure 16a.	Superficial
Pronator quadratus	Primary pronator of the forearm, stabilizes radius and ulna during pronation and weight-bearing.	Deep
<i>2. Muscles that flex and abduct or adduct the wrist</i>		
Flexor carpi radialis	Wrist flexion and abduction	Superficial
Palmaris longus	Weak wrist flexion and palmar aponeurosis tension.	Superficial
Flexor carpi ulnaris	Wrist flexion and adduction.	Superficial
<i>3. Muscles that flex the medial four fingers</i>		
Flexor digitorum superficialis	Flexes the wrist, the metacarpophalangeal (MCP) joint, and the proximal interphalangeal (PIP) joints of the digit 2-5. It can flex each finger independently.	Intermediate
Flexor digitorum profundus	Only muscle that flexes the distal phalanges 2-5 at DIP joints, helps make a full fist, and assists wrist flexion.	Deep
<i>4. Muscles that flex the thumb</i>		
Flexor pollicis longus	Flexes all joints(interphalangeal (IP), MCP, and carpometacarpal (CMC)) of the 1st digit (thumb). Assists wrist flexion.	Deep

Table 6: Flexor and pronator muscles of the forearm and their primary function [16]

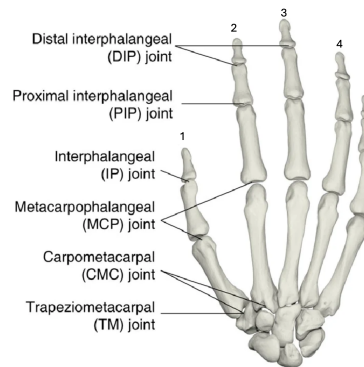


Figure 14: Joints of the right hand, shown in dorsal view [107].

it functions primarily as an elbow flexor and is therefore placed in 6, for the flexor muscles of the forearm. Similarly, the extensor muscles can be categorized according to their roles in wrist extension, finger extension, thumb movement, and forearm supination, as shown in Table 7. The muscles in the right forearm are visualized in Figure 15, where Figure 15a shows the anterior side, and 15b shows the posterior side of the forearm. The joints in the hand are visualized in Figure 14.

Muscle	Primary Function	Layer
<i>1. Muscles that extend and abduct or adduct the wrist</i>		
Extensor carpi radialis longus	Extend and abduct the hand at the wrist. Active during fist clenching.	Superficial
Extensor carpi radialis brevis	Extend and abduct the hand at the wrist. The primary extensor during grasping.	Superficial
Extensor carpi ulnaris	Extends and adducts the wrist. Active during fist clenching. Stabilizes the wrist during strong grip.	Superficial
<i>2. Muscles that extend the medial four fingers</i>		
Extensor digitorum	The primary extensor of digits 2-5, where it extends the metacarpophalangeal(MCP) joint, proximal interphalangeal (PIP) joint and distale interphalangeal (DIP). It assist wrist extension.	Superficial
Extensor digiti minimi	Extends the 5th digit primarily at MCP, secondary at the IP. Assists wrist extension.	Superficial
Extensor indicis	Extends the index finger independently. Assists ED extending the index finger. Assists wrist flexion.	Deep
<i>3. Muscles that extend or abduct the thumb</i>		
Abductor pollicis longus	Abducts the thumb and extends it at CMC joint. Assists wrist extension.	Superficial
Extensor pollicis longus	Extends the distal phalanx of thumb at IP joint, extends MCP and carpometacarpal (CMC) joints.	Deep
Extensor pollicis brevis	Extends the proximal phalanx of thumb at MCP joint; assists CMC joint extension.	Deep
<i>4. Muscles that supinate the forearm</i>		
Supinator	Supinates the forearm. Main muscle for slow, unresisted supination.	Deep

Table 7: Extensor and supinator muscles of the forearm and their primary functions [16]

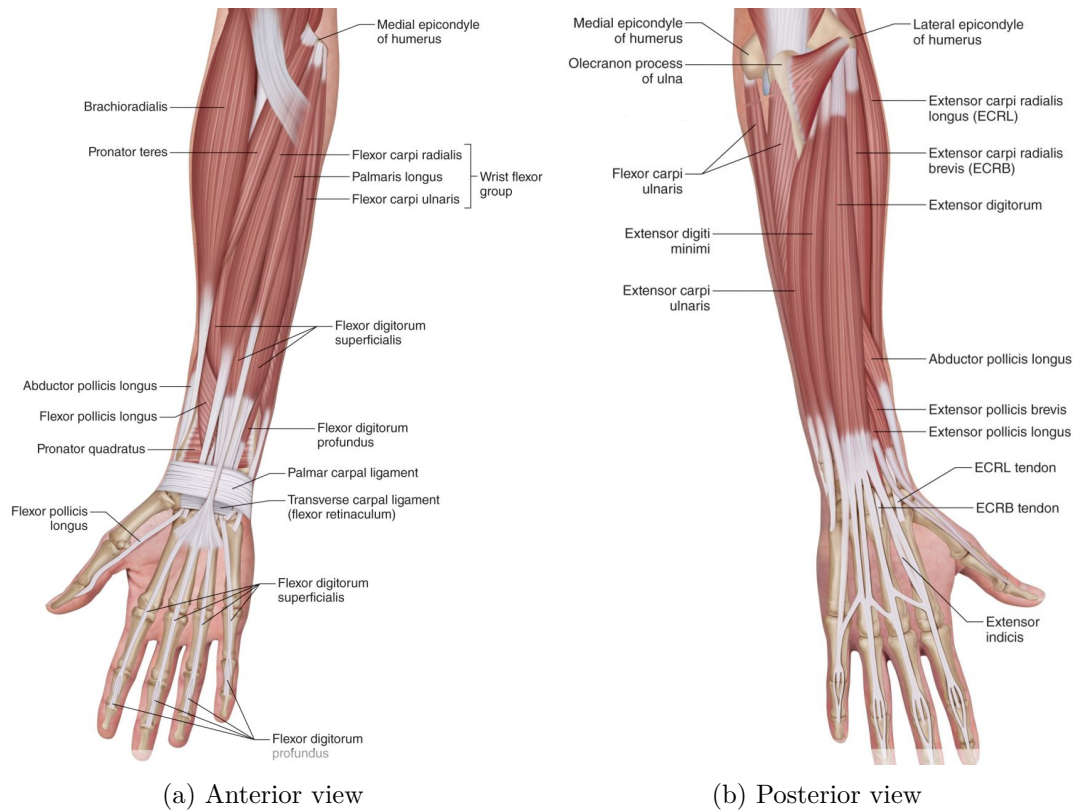


Figure 15: Muscles of the right forearm, a) shows the forearm from the anterior view and b) shows the forearm from the posterior view [108].

A.6.3 Extrinsic Muscle Activity During Essential Hand Movement

The extrinsic forearm muscles work together in coordinated activation patterns to perform functional wrist, hand and finger movements. During ADL, several muscles are activated simultaneously to generate movement while also stabilizing the wrist and hand [16, 109]. The following section describes the primary extrinsic muscles involved in: forearm rotation, wrist flexion, wrist extension, hand opening, and hand closing.

Pronation and supination are essential for a large range of ADL tasks, the movements enabling the forearm to supinate or pronate [110]. The wrist is involved in many functional activities, and functional wrist movement is essential for performing ADL activities [111]. Supination and pronation of the hand, and flexion and extension of the wrist, are essential movements of the hand in ADL and are therefore chosen as essential movements in this project.

Movements from a pronated position to the edge position, see figure 16a, with flexed elbow require coordinated activity between the pronator teres, pronator quadratus, supinator muscles, and the brachioradialis. The pronator teres and pronator quadratus contribute to forearm pronation, while the supinator rotates the forearm into supination [110]. The brachioradialis contributes to keeping the flexed elbow position [16, 106]

Wrist flexion is mainly generated by the flexor carpi radialis, palmaris longus, and flexor carpi ulnaris and supported by the flexor digitorum superficialis [111, 16]. These flex the wrist while contributing to wrist stabilization during grasping tasks [111, 16].

Wrist extension is primarily produced by the extensor carpi radialis longus, extensor carpi radialis brevis, and extensor carpi ulnaris [111, 16]. These muscles are also important wrist stabilizers during grip activities. Furthermore, the extensor digitorum additionally, assists wrist extension [111, 16]. Hand opening primarily involves extension of the finger and the thumb. Finger extension is mainly generated by the extensor digitorum, extensor indicis, and the extensor digiti minimi. Thumb opening and abduction involve coordinated activation of the abductor

pollicis longus, extensor pollicis brevis. These muscles are important for preparing the hand for grasping tasks and releasing objects.

Hand closing and grasp formation depend primarily on activation of the finger flexor muscles. The flexor digitorum superficialis flexes the proximal interphalangeal joints of the digits 2-5, while the flexor digitorum profundus flexes the distal interphalangeal joints, enabling formation of a full fist. Thumb flexion during grasp is mainly produced by the flexor pollicis longus, which performs grip strength during power grasping.

The study from Jarque-Bou(2024), which analyzed muscle activation during different grip variations, emphasized that during the cylindrical grip, the flexor digitorum profundus, flexor digitorum profundus, and the flexor pollicis longus were the most active muscles compared to other grips [109].

In figure 16, the described hand movements are visualized.

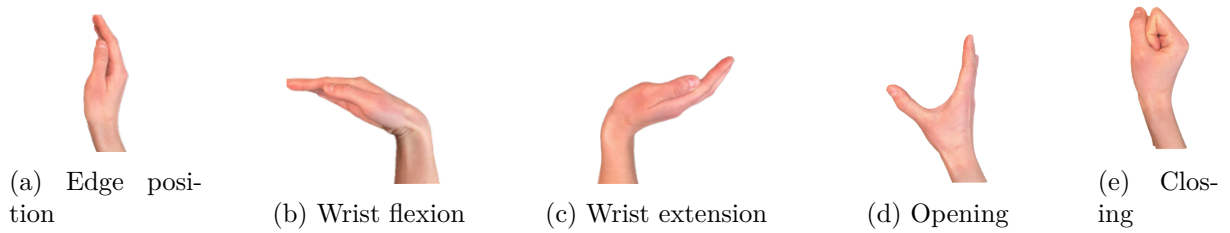


Figure 16: Visualization of the described hand movements.

A.6.4 Alterations in Muscle Coordination and Hand Function After Stroke

Coordinated activation between flexor and extensor muscle groups is essential for normal hand function [16]. During normal movement, agonist muscles are activated while antagonist muscles are simultaneously inhibited, allowing smooth and controlled wrist and finger movements [16]. Wrist and finger extension requires activation of the extensor muscles while the flexor muscles remain relatively relaxed, whereas grasping and hand closing require the opposite activation pattern. This is necessary for efficient motor control and coordinated hand movements [16].

Kuo and Hu (2018) describe spasticity as a common condition after stroke, that affects the flexor muscles of the upper limb. The increased flexor activity contributes to the characteristic flexed wrist and clenched hand posture frequently observed in stroke patients and leads to impaired hand function and limitations in ADL [18]. Since the flexor muscles generally possess greater force-generating capacity than the extensor muscles [17, 16], impaired neural control following stroke often causes movement patterns to become dominated by flexor activity [17].

Kamper and Rymer (2001) also emphasized that the disruption in the activation patterns after stroke results in abnormal co-activation between flexor and extensor muscles, which leads to impaired wrist control and reduced hand opening ability. They demonstrated that stroke patients had increased activation of the flexor muscles together with insufficient activation of the extensor muscles during attempted extending movements, such as wrist and finger extension. Indicating an impaired activation ratio between flexor and extensor muscle groups. The study further showed difficulty producing extension movement during isometric tasks, suggesting that impaired movement control, rather than spasticity alone, contributes to the abnormal movement patterns observed after stroke. A more recent study by Tacca et al.(2024) emphasizes that this co-activation is a common post-stroke symptom.

These studies show that altered muscle activation patterns impair grasp release, hand opening, and coordinated wrist movements, making rehabilitation of the extensor muscles and restoration of balanced muscle activation important targets in post-stroke upper limb therapy.

A.7 Measuring Forearm Muscle Activity Alterations with Electromyography

Electromyography (EMG) is a technique used to record the electrical activity derived by the muscles during contraction [19]. The EMG signal reflects the summation of action potentials generated by active motor units, which can be detected non-invasively at the skin surface, providing a window into neuromuscular activity, valuable for clinical assessment and rehabilitation [19]. In stroke rehabilitation of upper limb, biofeedback through surface EMG (sEMG) as an assessment to evaluate muscle activation is widely used to enhance motor function [20].

The sEMG reflects muscle activity and post-stroke physiological changes at the skin surface [112, 22]. Signals can be recorded even when patients are unable to complete their intended movements [112]. Multi-channel recordings make it possible to capture activation patterns across a muscle group simultaneously, enabling recognition of movement intent [112, 22]. To improve neuromuscular rehabilitation, sEMG has been utilized to assess muscle activation, while it is suitable for the detection of muscle co-activation in monitoring upper limb improvements [21]. Overactivation in the antagonist muscles during upper limb co-contraction is a common post-stroke symptom measurable with EMG on the forearm flexors and extensors [22].

In context to sEMG, High-density sEMG (HD-sEMG) is a promising complement to traditional assessment methods, providing non-invasive, high-resolution recordings of neuromuscular activity [23, 22]. Compared to sEMG, it allows more detailed mapping of motor unit activity and coordination patterns, offering a more objective understanding of post-stroke motor impairments and supporting individualized rehabilitation [23, 22]. In a study by Tacca et al. (2024), movement coordination of the extrinsic hand muscles was characterized across a broad range of functional movements using a wearable HD-EMG sleeve spanning the forearm. The study shows that the HD-EMG sleeve can identify detailed patterns of muscle activity, even in people who are unable to perform hand movements fully physically. Furthermore, the study emphasized the potential for detecting altered motor control post-stroke through the use of HD-EMG-derived features, contributing to better assessment and more targeted rehabilitation.

The growing interest in wearable devices for continuous electrophysiological monitoring has increased the need for more comfortable solutions, mitigating the barriers to adoption [113]. Smart textiles made with conductive materials offer a promising alternative for long-term monitoring [113]. Traditional “wet” electrodes rely on a conductive gel for signal transfer, but the gel dries out over time, reducing signal quality, making it unsuitable for long-term wearable use [113, 114]. Textile electrodes are “dry”, eliminating the issue of electrode dehydration and better suited for long-term use [113]. By using textile sleeves, it enables faster placement of the electrodes and possibilities to independently place the electrodes, especially for individuals with impaired hand function [115].

B System Design

The system design is based on the problem analysis presented in Appendix A and the expert interview documented in the Appendix E of the occupational therapist with 40 years of experience in the neuro-rehabilitation field. The problem analysis identified the key challenges inherent to stroke rehabilitation, including the complexity of motor and cognitive recovery, the need for repetitive and personalized training, and the importance of maintaining patient motivation over long recovery periods. The expert interview complemented these findings by providing practical and clinical perspectives on how these challenges manifest in rehabilitation practice, and which exercises and design considerations are most relevant to upper limb recovery, ensuring that the resulting system design is clinically grounded and aligned with real-world rehabilitation needs.

In this work, the requirements are derived from the problem analysis and expert interview, and prioritized using the MoSCoW model [56]. The Must-have requirements represent the highest priority, followed by Should-have, Could-have, and Won't-have. This work prioritizes to implement the Must-have requirements to ensure that the most important features and functionalities are developed before implementing lower-prioritized requirements. Lower-priority user requirements focus on further improvement of the system but are not necessary for the system to exist.

B.1 User Scenario

The system is designed to be used by a single patient assisted by a therapist. The patient performs self-directed exercise combinations. Therefore, the system must support their motivation by nudging the patient to complete the selected exercise combinations. Before using the system, a therapist assists the patient in attaching and connecting to the NeuraLoop device. Once the system is set up, the therapist selects the exercise combinations in the system before starting the training session. Prior to the training session, a calibration phase assisted by the therapist is performed, during which the system adapts to the patient's ability level.

During the training session, the patient attempts to perform different exercise combinations to support the rehabilitation of hand function. The exercise combinations of hand movements (Figure 16) are listed below:

- Movements from edge position to wrist flexion, and from edge position to wrist extension (EFE).
- Movements from edge position to wrist flexion, from edge position to wrist extension, and from edge position to a closed hand (EFEC).
- Movements from edge position to wrist flexion, from edge position to wrist extension, and from edge position to an open hand (EFEO).
- Movements from edge position to wrist flexion, from edge position to wrist extension, from edge position to a closed hand, and from edge position to an open hand (EFEEO).

The system detects whether the patient's attempted movements are performed correctly, and only correct execution allows the patient to progress past the obstacles. As the patient moves through the training session, they complete repetitions of the selected exercise combinations. A progression bar visually indicates how far the patient has progressed in the training session.

The requirements described in the following sections focus on the software.

B.2 User Requirements

The user requirements have been constructed based on a qualitative interview with a therapist at a rehabilitation center, described in appendix E and studies exploring EMG-controlled gamification in rehabilitation of hand function after stroke [37, 26, 31, 32, 33, 34, 35]. In the qualitative

interview, the therapist highlighted that many patients struggle with thumb activation and with controlling the arm in the edge position, yet these functions are crucial for reaching and grasping objects in everyday life. Therefore, training with edge position and open/close hand exercises is highly relevant to ADL rehabilitation. The rehabilitation plan is customized for each patient, therefore, it is important that the system can adjust exercises to the patient’s individual ability level. Additionally, the therapist emphasized that most patients have cognitive challenges, typically limited memory span and attention disorders. Therefore, it is crucial that the training exercises are simple and intuitive, and that the patient clearly understands how to perform each movement. Furthermore, the system should support maintaining the patient’s attention during the training session. Since patients are also responsible for completing their own training and reaching the required number of repetitions, the system should support the patient’s motivation and adherence.

Furthermore, studies highlight the importance of repetitive training of the wrist flexion and extension following stroke [52, 53, 54]. Wrist flexion and extension exercises play a central role in post-stroke upper limb rehabilitation, as targeted training of wrist flexor and extensor muscles has been shown to improve motor control and muscle activation [55, 53]. Particularly, wrist extension is essential for effective grasping and object manipulation, as it influences grip strength and hand function [53]. Repetitive task-specific training of wrist movements has been associated with improved motor performance and functional recovery [52, 54].

According to the qualitative interview, the system has two primary users: the therapist and the patient. The therapist is responsible for configuring the system by attaching the electrode sleeve and the NeuraLoop device and initiating the training session, while the patient is responsible for performing the exercises in the training session. The user requirements are listed in Table 8.

Table 8: User requirements with MoSCoW prioritization

ID	User Requirements	MoSCoW
UR1	The therapist must have an interface to set up training session with training dosage and exercise combinations consisting of edge position, wrist flexion, wrist extension, close and open hand.	Must
UR2	The patient must receive support from the system to train the edge position.	Must
UR3	The patient must receive support from the system to train wrist flexion.	Must
UR4	The patient must receive support from the system to train wrist extension.	Must
UR5	The patient must receive support from the system to train to closing the hand.	Should
UR6	The patient must receive support from the system to train to opening the hand.	Should
UR7	The patient must be informed how to perform edge position, wrist flexion, wrist extension, closing and opening of the hand to complete the exercises.	Must
UR8	If the patient is inactive for more than 30 seconds, the system must remind the patient.	Could
UR9	The patient must be informed about the number of repetitions at the end of the session.	Could

We designed a gamified rehabilitation environment in which the patient controls a rocket in a game world using hand movement intentions inferred from forearm muscle activity recorded via

HD-EMG. The signals are processed in real time by an embedded microcontroller unit (MCU) in the NeuraLoop device, which classifies specific hand gestures. These gestures are mapped to in-game actions, enabling the user to navigate the rocket through the environment with continuous visual feedback supporting engagement and rehabilitation. The actions associated with the attempted hand movements are listed in Table 9.

Movement	Action
Edge position	Rocket moves forward
Wrist flexion	Rocket moves to the left
Wrist extension	Rocket moves to the right
Closed hand	Rocket shoot
Opened hand	Rocket increase speed

Table 9: Movement actions

B.3 System Requirements

The system requirements are divided into requirements derived from user requirements and requirements not derived from user requirements. The system is constructed to a right hand user. Therefore, the system requirements are developed for the right-hand user.

In table 10, the system requirements that are not directly derived from user requirements are presented. These requirements describe essential conditions that must be fulfilled in order to satisfy the user requirements. The identified conditions include an user-friendly interface of the application, connecting to NeuraLoop and calibration of muscle activity. The interface of the application must be intuitive and easy to use to ensure an effective user experience. The system depends on the NeuraLoop, therefore, connecting to NeuraLoop is necessary to use the system. Calibration of muscle activity is required as it is directly related to the algorithm used to detect hand movements during training sessions. Consequently, a calibration phase is essential to ensure accurate detection and system performance.

Testing of terms in bold will be described further after Table 10.

Table 10: System requirements not derived from the user requirements with MoSCoW prioritization.

ID	System Requirements	MoSCoW
SR0.1	The system must be an application that can be started by clicking or tapping it.	Must
SR0.2	The system must perform signal processing and gesture classification on the NeuraLoop device microcontroller.	Must
SR0.3	The system must show a guide to connect with NeuraLoop.	Must
SR0.4	The system must be able to connect with NeuraLoop.	Must
SR0.5	The system must show whether the connection to the NeuraLoop is established.	Must
SR0.6	The system must be able to adapt to the muscle activity level of the user through a calibration before starting the training session.	Must
SR0.7	The system must contain a function that starts the training session when activated	Must

Continued on next page

Table 10: System requirements not derived from the user requirements with MoSCoW prioritization. (Continued)

ID	System Requirements	MoSCoW
SR0.8	The system must track and show the completed training dosage and calculate the remain training dosage during training session.	Should
SR0.9	The system must show an option to end the training session before, the selected training dosage completed.	Could
SR0.10	The system's latency time from performing the movement to give visual feedback on the movement must be kept below 300 ms.	Should

In table 11, the system requirements derived from user requirements are presented. For each user requirement, a set of corresponding system requirements are defined to ensure that the user requirement is fulfilled.

Table 11: System requirements derived from the user requirements with MoSCoW prioritization.

ID	System Requirements	MoSCoW
SR1.1	The system must provide an interface where the therapist can select the exercise combination: EFE, EFEC, EFEO, EFECO	Must
SR1.2	The system must provide an interface where the therapist can select the training dosage, and the system must adjust the training session according to the selected dosage and end the session once the selected dosage is completed.	Should
SR2.1	The system must be able to detect the resting position.	Must
SR2.2	The system must be able to detect the edge position.	Must
SR2.3	The system must start the training session and move the rocket forward when the edge position is detected.	Must
SR2.4	The system must pause the training session when the movement from edge position to the rest position is detected.	Must
SR3.1	The system must be able to detect wrist flexion when wrist flexion is performed or attempted to be performed	Must
SR3.2	In the movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the left when the wrist flexion is detected.	Must
SR4.1	The system must be able to detect wrist extension, when wrist extension is performed or attempted to be performed.	Must
SR4.2	In movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the right when the wrist extension is detected.	Must
SR5.1	The system must be able to detect close hand when the closing hand movement is performed or is attempted to be performed.	Should
SR5.2	In movement combinations EFEC, EFECO, the enemies must spawn randomly.	Should
SR5.3	In movement combinations EFEC, EFECO, the rocket must shoot when the close hand is detected.	Should

Continued on next page

Table 11: System requirements derived from the user requirements with MoSCoW prioritization.
(Continued)

ID	System Requirements	MoSCoW
SR5.4	In movement combinations EFEC, EFECO, the enemy must die when hit by the shots from the rocket.	Should
SR6.1	The system must be able to detect an open hand when the hand opening movement is performed or is attempted to be performed.	Should
SR6.2	In movement combinations EFEO, EFECO, the rocket's speed must decrease over distance.	Should
SR6.3	In movement combinations EFEO, EFECO, the rocket must increase speed when close hand is detected.	Should
SR6.4	In movement combinations EFEO, EFECO, the first 5 seconds, when back in edge position after the open hand is detected, the rocket must keep the acquired speed level.	Could
SR7.1	In the movement combinations EFE, EFEC, EFEO, EFECO, the system must show a visualization of the hand movements wrist flexion, wrist extension, edge position, closing hand, and opening hand parred with the corresponding rocket actions.	Must
SR7.2	If paused by detecting resting position, the system must show a visualization of the edge position to start the training session and move the rocket forward.	Should
SR8.1	The system must detect if the user has been inactive in the training session for more than 30 seconds.	Could
SR8.2	If the user is detected inactive, the system must give audio feedback that recreates the users attention to the training session.	Could
SR8.3	If the user is detected inactive, the system must give visual feedback that recreates the users attention to the training session.	Could
SR9.1	The system must inform the user of the achieved number of repetitions when the training session is ended.	Could

B.4 Mock-Up

To illustrate how the system fulfills its requirements, a mock-up was developed representing the application as it would appear to the user. Once started, the user is presented with a start screen from which they can select between four exercise combination described in Section B.1. Each exercise combination maps to a corresponding in-game rocket action referred to as play mode, as listed below:

- EFE mapped to Side to Side
- EFEC mapped to Side to Side + Shoot
- EFEO mapped to Side to Side + Power
- EFECO mapped to Side to Side + Power + Shoot

The initial start screen is shown in Figure 17a and the start screen when all settings are completed is shown in Figure 17b. At the top of the start screen, a guide to connect to the NeuraLoop is shown. The box with the connection guide is red when the connection to the NeuraLoop is not established and contains a text at the top of the box that informs the user

about the missing connection with the text *NeuraLoop is not connected!*. When the connection to the NeuraLoop is established, the box changes color to dark gray, and the text changes to *NeuraLoop is connected!*. Below the connection box, a play mode selection box is placed. In this box, the target hand and the play mode must be selected. Beside the selection of play modes, the hand movements to be trained are visualized. The pictures inform the user which exercises are included in the play mode. When the user has selected play mode, the box will turn dark gray. When both the connection to the NeuraLoop is established, and the user has chosen the play mode, the calibration button will turn red. The calibration is not a part of the game and will not be shown in this Mock-Up, as the presented Mock-Up is focused on the game. When the calibration is completed, the lower box in the start screen with the text *Play* will turn green to inform the user that the game is ready to play. By pressing the play button, the game starts.

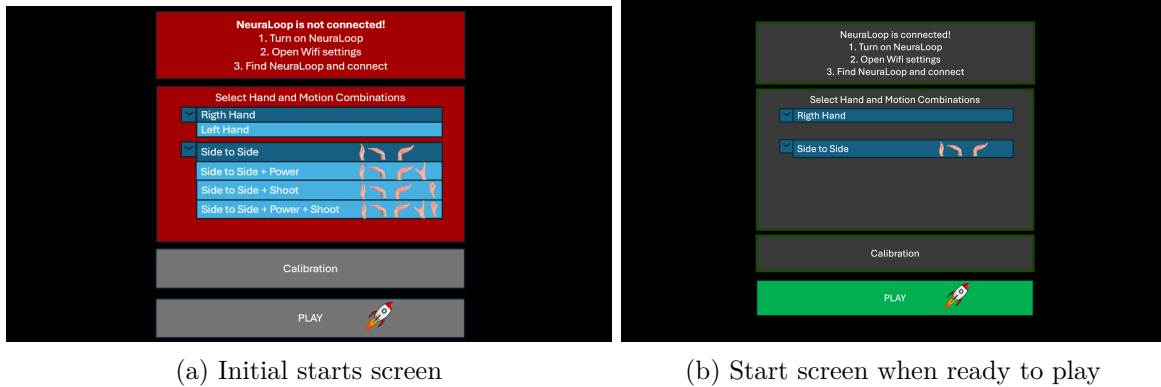


Figure 17: The Start Screen of the system contains multiple boxes, each is responsible for its own functionalities as NeuraLoop connection, exercise selection, calibration and play, see figure 17a. When the functionality is available, the box is red, except the play button, which turns green, see figure 17b. The boxes must be completed in sequence. Light gray boxes are not yet available, while dark gray boxes indicate completed boxes.

To play, the hand must be placed in the edge position. If the hand is not in the edge position for 30 seconds, the pause screen (see Figure 18) is displayed. In the pause screen, it is possible to continue the training session, or end the training session and the game provides audio feedback to redirect the user's attention to the training session.

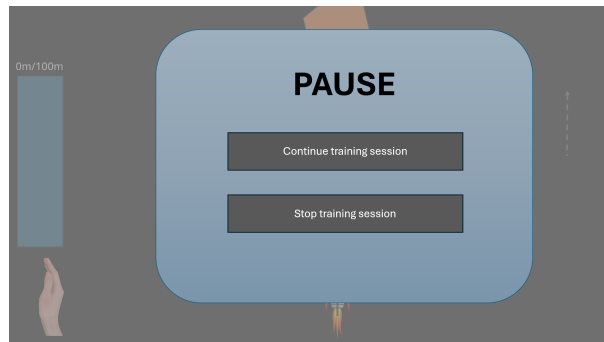
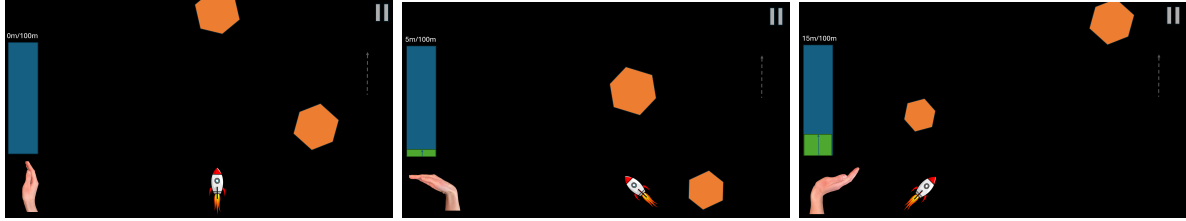


Figure 18: Pause Screen is shown when the pause button in the game is pressed or when the target arm is not in edge position.

Play mode *Side to side* is shown in Figure 19. The goal of the game is to fly the rocket a preset distance of 100m. A progress bar on the left side informs how close the rocket is to reach the goal. The rocket can do three actions: move forward, move left, and move right. In this play mode, the rocket automatically moves forward at a constant speed, shown by the rocket

always producing engine thrust fire, see figure 19a. Through the game, the rocket must avoid incoming asteroids by moving left or right. The rocket moves left by flexing the wrist and right by extending the wrist. When an asteroid is directed towards the rocket, an animation of the suggested movement will appear under the progression bar, see figure 19b and 19c. When 100m is reached, the game is completed.



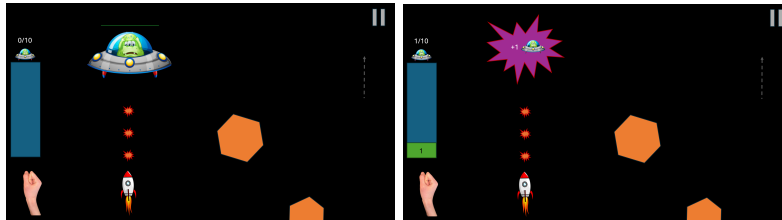
(a) Rocket moves forward

(b) Rocket moves left

(c) Rocket moves right

Figure 19: Play mode: Side to Side, in this play mode, the rocket can move a) left, b) forward, and c) right to avoid incoming asteroids

Play mode *Side to Side + Shoot* is an extension of play mode *side to side*, but here the goal is to shoot 10 enemies throughout the game, this is shown through a progression bar counting the defeated enemies. The rocket moves as described in play mode *side to side*, but in this game mode, enemies that shoot after the rocket will appear. To defeat an enemy, the rocket shoots the enemies, this is done by moving the rocket below the enemy and shoot by closing the hand, see figure 20a. When the enemy is hit, it will explode, and the progression bar on the left side of the screen will fill, see figure 20b. When 10 enemies are defeated, the game is completed.



(a) Rocket shoots enemy

(b) Enemy explodes

Figure 20: Play mode: Side to side + shoot, extension of play mode side to side. a) An enemy spawns in the game, and the rocket shoots at the enemy. b) The enemy is hit by the shot of the rocket and explodes.

Play mode *Side to Side + power* is also an extension of play mode *Side to Side*, but now the user must add power to the rocket for moving faster forward and prevent it from not slowing down. The goal of this play mode is to reach a preset distance of 100 meters, as in play mode *side to side*. In this play mode, the extended functionality is that the rocket's velocity will decrease with distance, to increase the speed of the rocket, the player must open the hand. In the beginning of the game the rocket start still or moving very slowly forward, this is shown in the speed bar beside the distance bar, and in that the rocket does not have engine thrust fire in the back, see figure 21a. To increase the speed of the rocket the player must open the hand, the longer time *open hand position* is hold, the more speed the rocket will gain until a maximum speed is reached. When the rocket is increasing speed it will be visualized by filling the speed bar, see figure 21b, and the engine thrust fire will be increased. When not adding speed to the rocket the rocket will slightly moving slower forward, this is shown by a decreasing speed bar and a decreasing or no engine thrust fire, see figure 21c.

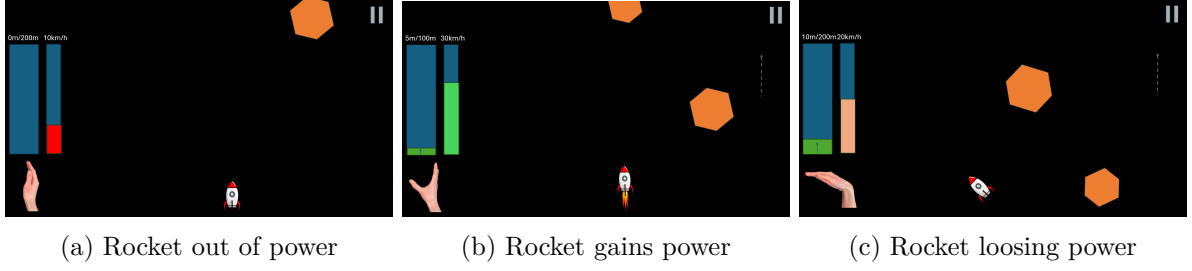


Figure 21: Play mode: Side to Side + Power, extension of play mode side to side. a) The rocket is out of power and moves very slowly, b) the rocket gains power and increases its speed by opening hand, c) the rocket slowly loses its speed during the game.

Play mode *side to side + shoot + power* is a combination of play mode side to side + shoot and play mode side to side + power. The goal of this game mode is to reach the preset distance of 100m. The rocket moves left and right by flexing and extending the wrist, see figure 19. It shoots by closing the hand as seen in figure 20, and it gains speed by opening the hand as seen in figure 21.

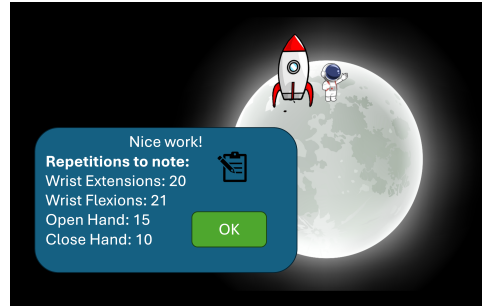


Figure 22: The end screen of the system shows the repetition performed in the training session.

When the goal of the play mode is reached, the game finishes, and the end screen will be shown, see figure 22, the end screen celebrate the users performance in the game by showing that the astronaut received the moon, and displaying the text *Nice work!*. Furthermore, the end screen reminds the user to manually note down the achieved movement repetitions in the game. Only the repetitions of the movement required for the play mode will be shown. The end screen also contains an *OK* button, which will lead the user back to the start screen when pressed.

B.5 System Workflow

The activity of the system workflow(Figure 3 in article) starts on *Calibrate for Muscle Activity Level*, here the fixed thresholds and values used for gesture recognition is being identified for the individual user of the system. Then the therapist *Opens the Application*. When the application is running, synchronization is used to split one incoming into multiple concurrent flows; *Connection to NeuraLoop*, *Choose Training Dosage*, and *Choose Movements* must be obtained before next activity which is *Play the Game*. In these activities the therapist are main responsible in collaboration with the patient. The game ends when *Selected training dosage Achieved*, which leads the user *Training Session Completed*, here the user but be informed the note the achieved training dosage in their personal training documents. From *Play the Game* an arrow to a more elaborated activity diagram for the game is shown.

C Test

In the following sections, the test protocol for the implemented system requirements is described.

C.1 Test of Connection to NeuraLoop and Game Application

Tested Requirements:

- SR0.1: The system must be an application that can be started by clicking or tapping it.
- SR0.3: The system must show a guide to connect with NeuraLoop.
- SR0.4: The system must be able to connect with NeuraLoop.
- SR0.5: The system must show whether the connection to the NeuraLoop is established.
- SR0.7: The system must contain a button that starts the training session when activated.
- SR0.8: The system must track and show the completed training dosage and calculate the remaining training dosage during training session.
- SR1.1: The system must provide an interface where the therapist can select the exercise combination: EFE, EFEC, EFEO, EFECO.
- SR1.2: The system must provide an interface where the therapist can select the training dosage, and the system must adjust the training session according to the selected dosage and end the session once the selected dosage is completed
- SR7.1: In the movement combinations EFE, EFEC, EFEO, EFECO, the system must show a visualization of the hand movements, wrist flexion, wrist extension, edge position, closing hand, and opening hand parred with the corresponding rocket actions.

Test Conditions: The system has been build as with the bash script and the installation.sh file has been executed. In the system, only wrist flexion and wrist extension are implemented. Therefore, only wrist flexion, wrist extension, and rest position corresponding to edge position hand movements must be visualized. The training dosage distance is in meters, where the rocket is moving with a speed of 1 meter per second. In the system, EFE exercise combination is implemented. Therefore, the test is focused on the selection of EFE exercise combination. The user must have performed the calibration and have the NeuraLoop system attached to the right arm and have the output from the calibrations, three flexion channels, three extension channels, base threshold and hysteresis threshold, before starting the test. The system is tested on a Linux Computer with Operational System: Ubuntu 24.04.4 LTS.

Test protocol:

The protocol must be repeated 5 times, each time with a different training dosage value, the different training dosage values must be: 10, 20, 30, 40, 50.

1. Ensure that the computer is not connected to the NeuraLoop WiFi, and the NeuraLoop is turned OFF.
2. Start the game application: **SpaceGame** by clicking the application icon, and turn on the NeuraLoop.
3. Enter the output value from the Calibration, 3 channels for extension, 3 channels for flexion, base threshold and hysteresis threshold. The values must be entered in the belong input boxes in the upper box for calibration output values.
4. Verify that a guide for connecting to NeuraLoop is shown, and follow the guide to establish the WiFi connection to the NeuraLoop.

5. Verify that the computer is connected to the NeuraLoop WiFi, and that the connection status field in the application changes color to green.
6. Verify that the start screen consists of a scroll bar with the exercise combination **Side to Side**, **Side to Side + Shoot**, **Side to Side + Power**, and **Side to Side + Shoot + Power**, and a scroll bar with **Right Hand** and **Left Hand**, select the **Side to Side** and **Right Hand**.
7. Verify that the application start screen consists of a field where the training dosage could be changed, and enter training dosage for the test repetition, in the field.
8. Verify that the application start screen consists of a **Play Button**, and start the training session by clicking the **Play Button**. Meanwhile the **Play Button** is pressed a timer on an external devices must be started.
9. During the training session:
 - Verify that visualization of wrist flexion, wrist extension, and edge position is visualized in the bottom of the screen, below the rocket, and that the gestures are parred with its corresponding movement of the rocket.
 - Verify that the training session environment consists of a progression bar, and that the progression bar is changing when the rocket is moving forward.
10. Verify that the end screen appears when the training session is over and stop the timer on the external device immediately.
11. Press **Back to Menu Button**, verify that the main menu appears and that the NeuraLoop is still connected
12. Disconnect the connection to the NeuraLoop in the game and verify that NeuraLoop is flashing green, then disconnect the NeuraLoop WiFi connection on the computer, and turn the NeuraLoop OFF.
13. Verify that the connection status field in the application changes color to red.
14. Close the application by tapping the Windows button on the keyboard and then clicking the cross in the upper right corner of the application.

Success Criteria:

- SR0.1: The application opens successfully 5/5 times.
- SR0.3: The system shows a guide to connect with NeuraLoop, and the connection can be established by following the guide.
- SR0.4: The system connect to the NeuraLoop 5/5 times.
- SR0.5: The system shows whether the connection to the NeuraLoop is established 5/5 times.
- SR0.7: The **Play Button** start the training session when activated 5/5 times.
- SR0.8: The system tracked and showed the completed training dosage and calculated the remaining training dosage during the training session by filling the progress bar continuously 5/5 times.
- SR1.1: The system provided an interface to select the exercise combination, and the EFE exercise combination could be selected 5/5 times.

- SR1.2: The system provided an interface where the training dosage could be entered, adjusted the training session according to the selected dosage, and ended the session within 3 seconds once the selected dosage was completed 5/5 times.
- SR7.1: The system showed a visualization of the hand movements wrist flexion, wrist extension, and edge position paired with their corresponding rocket actions 5/5 times.

C.2 Test of Microcontroller Processing

Tested Requirement:

- SR0.2: The system must perform signal processing and gesture classification on the NeuraLoop device microcontroller.

Test Conditions:

The user must have performed the calibration and saved the recording.

Test Protocol:

1. Upload the saved calibration recording in MATLAB.
2. Plot the differential signal alongside the gesture classification output, with markers indicating when the entry and exit thresholds are crossed.
3. Verify that the gesture labels produced by the MCU classification correspond correctly to the threshold crossings in the differential signal.
4. Repeat 5 times with different calibration recordings.

Success Criteria:

- The gesture byte changes when the signal exceeds or falls below the entry threshold successfully 5/5 times.

C.3 Test of Calibration

Tested Requirements:

- SR0.6: The system must be able to adapt to the muscle activity level of the user through a calibration before starting the training session.

Test Conditions:

In the system, only wrist flexion and wrist extension are implemented. Therefore, the calibration consists of wrist flexion, wrist extension and rest position corresponding to edge position. Calibration is conditional on the patient having the NeuraLoop system attached to the right arm. The calibration protocol is described in the article.

Test Protocol:

1. Open the EMGCalibrationTool.m file in MATLAB, turn on the NeuraLoop, and establish the WiFi connection to the NeuraLoop.
2. Run the EMGCalibrationTool.m file in MATLAB, and record a calibration recording where the user follows the calibration protocol.
3. Verify that the calibration script is given three channels for flexion and three channels for extension, and that exit threshold and hysteresis threshold are given. Save the calibration output from the Calibration window.
4. Disconnect the NeuraLoop WiFi connection and turn off the NeuraLoop.

5. Close the Calibration window.
6. Repeat the step 1-5, 5 times.

Success Criteria:

- SR0.6: The calibration results correctly visualized three flexion and three extension channels in a plot together with the corresponding thresholds. The classification accuracies for the edge position, wrist flexion, and wrist extension > 70%.

C.4 Test of Total System Latency Time

Tested Requirement:

- SR0.10: The system's latency time from performing the movement to give visual feedback on the movement must be kept below 300 ms.

Test Conditions:

The Test of Gesture recognition has been filmed, and there exists a video of a user doing the test where the hand and the gamescreen are clearly visible

Test Protocol:

1. Upload the video recorded doing the Test of Gesture recognition to the VideoLatency.py script.
2. Run the script and navigate through the video by pressing the key "d" for next frame and "a" for previous frame.
3. For each player action, gesture onset must be noted by pressing "g", gesture end must be noted by pressing "h", and the rockets first pixel movement must be noted by pressing "r". When both gesture onset, gesture end, and rocket reaction are noted for a player action, "s" must be pressed, to save the results for the player action.
4. When no more player actions and the video has run to the end. Quit the video by pressing "q".
5. Run the "latency.ipynb" Python Notebook script, and note the results.

Success Criteria:

1. SR0.10: Time from gesture completion to rocket movement must not exceed 300ms, evaluated over 5 flexion and 5 extension transitions.

C.5 Test of Gesture Recognition

Tested Requirements:

- SR2.2: The system must be able to detect the edge position.
- SR3.1: The system must be able to detect wrist flexion when wrist flexion is performed or attempted to be performed.
- SR3.2: In the movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the left when the wrist flexion is detected.
- SR4.1: The system must be able to detect wrist extension, when wrist extension is performed or attempted to be performed.

- SR4.2: In movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the right when the wrist extension is detected.
- SR9.1: The system must inform the user of the achieved number of repetitions corresponding to the detected gestures when the training session is ended.

Test Conditions:

In the system, only wrist flexion and wrist extension are implemented. Therefore, only wrist flexion, wrist extension, and rest position corresponding to edge position must be performed corresponding to the EFE exercise which is named Side to Side in the game. The user must have performed the calibration and having the NeuraLoop system attached to the right arm and have the output from the calibrations, three flexion channels, three extension channels, base threshold and hysteresis threshold, before starting the test. Further more the test must be filmed for further latency testing. It is important that the user's hand that is wearing the NeuraLoop system and the gamescreen are clearly visible.

Test Protocol:

1. Start the game in Unity.
2. Enter the three flexion channels, three extension channels and the thresholds from the calibration.
3. Verify that the computer is connected to the NeuraLoop WiFi, and that the connection status field in the application changes color to green after following the guide for connection.
4. On the start screen, select the Side to Side and Right Hand and set the training dosage to 180.
5. Press the **Play Button** to start the training session.
6. During the training session:
 - Start in edge position and change to the position corresponding to the arrow highlighted in green in the training session and proceed to the training session is over.
7. Verify that the achieved number of repetitions is shown in the end screen when the training session is over.
8. Verify that the achieved number of repetitions matches the logged detected repetitions after the game period.
9. Disconnect the connection to the NeuraLoop in the game and verify that NeuraLoop is flashing green, then disconnect the NeuraLoop WiFi connection on the computer, and turn the NeuraLoop OFF.
10. Close the game in Unity.
11. Repeat the step 1-10, 2 times.

Success Criteria:

- SR2.2: Classification accuracy $> 70\%$. A detection is successful if when the interval the up arrow appears has only one logged edge position.
- SR3.1: Classification accuracy $> 70\%$ A detection is successful if when the interval the left arrow appears has only one logged wrist flexion.

- SR3.2: Manual verification of the rocket moving to the left when wrist flexion is detected during a 180-second training session, 2/2 times.
- SR4.1: Classification accuracy > 70%. A detection is successful if when the interval the right arrow appears has only one logged wrist extension.
- SR4.2: Manual verification of the rocket moving to the right when wrist extension is detected during a 180-second training session, 2/2 times.
- SR9.1 The system showed the achieved number of repetitions on the end screen matching the logged gestures.

C.6 Summarization of System Requirements Test

In following table 12, the test results of the system requirements is listed.

Table 12: System Requirements for Testing.

ID	System Requirements	MoSCoW	Result	Fulfilled
SR0.1	The system must be an application that can be started by clicking or tapping it.	Must	5/5	Pass
SR0.2	The system must perform signal processing and gesture classification on the NeuraLoop device microcontroller.	Must	5/5	Pass
SR0.3	The system must show a guide to connect with NeuraLoop.	Must	5/5	Pass
SR0.4	The system must be able to connect with NeuraLoop.	Must	5/5	Pass
SR0.5	The system must show whether the connection to the NeuraLoop is established.	Must	5/5	Pass
SR0.6	The system must be able to calibrate the movements consisting of edge position, wrist flexion, wrist extension, close and open hand before starting the training session.	Must	5/5	Pass
SR0.7	The system must contain a button that starts the training session when activated	Must	5/5	Pass
SR0.12	The system must track and show the completed training dosage and calculate the remain training dosage during training session.	Should	5/5	Pass
SR0.13	The system's latency time from performing the movement to give visual feedback on the movement must be kept below 300 ms	Should	10/10	Pass
SR1.1	The system must provide an interface where the therapist can select the exercise combination: EFE, EFEC, EFEO, EFECO	Must	5/5	Pass
SR1.2	The system must provide an interface where the therapist can select the training dosage.	Should	5/5	Pass

Continued on next page

Table 12: System Requirements for Testing. (Continued)

ID	System Requirements	MoSCoW	Result	Fulfilled
SR1.3	The system must adjust the training session depending on the selected training dosage	Should	5/5	Pass
SR2.2	The system must be able to detect the edge position	Must	74%	Pass
SR3.1	The system must be able to detect wrist flexion when wrist flexion is performed or attempted to be performed	Must	53%	Fail
SR3.2	In the movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the left when the wrist flexion is detected.	Must	2/2	Pass
SR4.1	The system must be able to detect wrist extension, when wrist extension is performed or attempted to be performed.	Must	91%	Pass
SR4.2	In movement combinations EFE, EFEC, EFEO, EFECO, the rocket must move to the right when the wrist extension is detected.	Must	2/2	Pass
SR7.1	In the movement combinations EFE, EFEC, EFEO, EFECO, the system must show a visualization of the hand movements wrist flexion, wrist extension, edge position, closing hand, and opening hand parred with the corresponding rocket actions.	Must	5/5	Pass
SR9.1	The system must inform the user of the achieved number of repetitions when the training session is ended and remind the user to document the repetitions.	Could	2/2	Pass

D Structured Literature Review

The purpose of the systematic literature search was to investigate the following research question: *How could EMG-based system integrated with a game contribute to hand rehabilitation in post-stroke patients?*

The research question aims to examine therapeutic contribution of game-based systems utilizing surface electromyography (sEMG) for hand rehabilitation after stroke. By addressing this question, the intention was to identify the current state of the art including rehabilitation approaches utilizing sEMG-controlled games. In the systematic literature search, PubMed, Scopus and IEEE Xplore were utilized. Based on these three databases, a broad and interdisciplinary range of literature was covered, including clinical, biological, and technological research relevant to stroke rehabilitation and sEMG-based game systems. PubMed is a comprehensive database in medicine and health sciences, providing access to biomedical literature [116]. Selecting PubMed ensures access to a strong foundation of clinically relevant and biologically grounded research. This is particularly important for identifying studies on stroke rehabilitation, muscle activation, and the use of sEMG, which are central to understanding the therapeutic potential of EMG-based game systems for hand rehabilitation. Scopus is a large multidisciplinary database covering scientific, technical, and medical literature. Its use ensures broad coverage of research on rehabilitation technologies, including game-based systems and sEMG applications [117]. IEEE Xplore is a database focused on engineering and technology research. It provides access to studies on EMG-based systems, signal processing, and game development, which are essential for understanding the technical aspects of sEMG-based rehabilitation [118].

D.1 Aspects for Search String

Based on the research question, five aspects have been identified for the database search: *Stroke*, *EMG*, *Hand*, *Rehabilitation*, and *Game*. Search terms are established using truncation and thesaurus terms in the databases. Furthermore, truncation is visualized with '*' to ensure that all suffixes of the words are included. The aspects are combined using AND in the search string

Block Search					
	AND				
	Aspect 1	Aspect 2	Aspect 3	Aspect 4	Aspect 5
OR	Stroke*	EMG	Hand*	Rehabilitation	Game*
	Post Stroke	Surface EMG	Upper Limb*	Therapy*	Gamification
	Hemipare*	Electromyogra*	Upper Extremit*	Recover*	Training Environment*
	Apoplex*	sEMG	Wrist	Training	Gamified
	Cerebrovascular Accident*	Muscle Activit*	Membrum superius	Exercise*	Videogame*
	CVA	Myoelectric interface*	Metacarpus		
	Brain Vascular Accident*	HD-EMG	Grasp*		
		High Density Electromyo- gra*	Grip*		

Table 13: Block search strategy

to ensure that all aspects are represented in retrieved articles. In each aspect, search terms are connected with OR to capture at least one search term from the aspect included in the articles. The search string is configured to identify occurrences of these terms in the title or abstract. A block search has been conducted based on these aspects and is visualized in Table 13.

D.2 Inclusion and Exclusion Criteria

The resulting articles based on the search string in the three databases was uploaded in Rayyan to screen the articles. Rayyan is a research collaboration platform that can reduce screening time by enabling multiple members to participate in the screening process [119]. Duplicates are identified by using Rayyans *possible duplicates* [120] to exclude articles which are $\geq 85\%$ similar. This cutoff is selected to ensure that highly similar studies which would not contribute additional to the review were excluded. Possible duplicates $\geq 85\%$ are compared manually based on title, abstract, publication year, and authors before exclusion.

The exclusion process was divided into two phases where articles were excluded based on their title/abstract and full text review. During this process, the inclusion and exclusion criteria in table 14 were utilized, where the exclusion criteria were focused on exclusion based on title/abstract, and the inclusion criteria were crucial during the full text review. The purpose of the criteria was to structure the search in accordance with answering the research question and to ensure reproducibility with a defined set of criteria, minimizing bias.

The systematic literature review and exclusion process conducted resulted in 16 articles that addressed the research question.

The summary of the systematic literature review is visualized in figure 23.

Inclusion criteria	Exclusion criteria
Articles addressing hand rehabilitation in adult stroke patients.	Articles addressing animal or pediatric studies.
Articles addressing game-based rehabilitation interventions.	Articles addressing electrical stimulation as the primary intervention.
Articles addressing the use of sEMG as measurement or control.	Articles addressing robotic systems, exoskeletons, orthoses, or robotic hands as the main intervention.
Articles addressing full-text peer-reviewed research.	Articles addressing lower-limb rehabilitation only.
	Articles addressing shoulder or proximal arm rehabilitation without including the hand.
	Articles addressing pharmacological interventions.

Table 14: Inclusion and exclusion criteria for the systematic literature review.

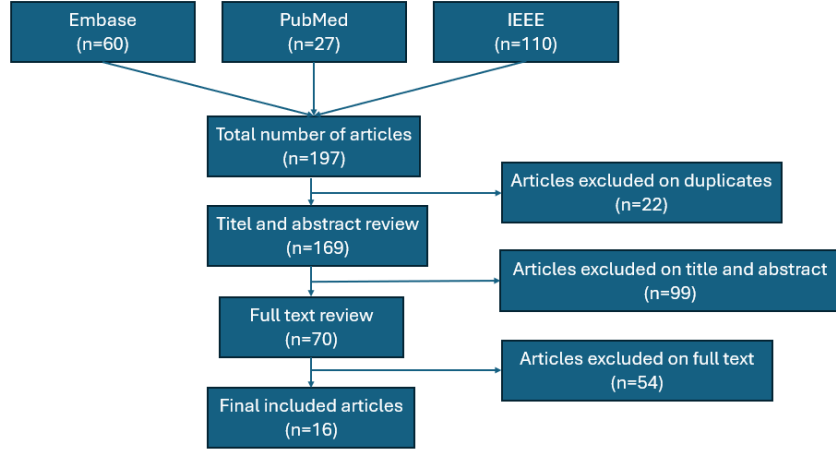


Figure 23: Visualization of the results after the structured literature review.

D.3 Summary of included articles

A summary of the included articles from structured literature review is presented in Table 15, including their methods, and key findings.

Table 15: Summary of included articles from the structured literature review.

Study	Methods	Findings
Marin-Pardo et al. (2024)	9 chronic stroke patients with moderate to severe upper extremity (FMA-UE = 26 ± 9.89) participated in a 6-week home-based telerehabilitation program (30 sessions, 1 hour each). The intervention used sEMG from wrist muscles combined with a VR serious game to provide real-time biofeedback for wrist flexion and extension training (TeleREINVENT).	Improvement in wrist extension and flexion. High adherence. Increased motivation and perceived quality of life.
Marin-Pardo et al. (2022)	5 chronic stroke patients with moderate to severe upper extremity (FMA-UE = 19-40) completing 30 EMG-based VR training sessions over 6 weeks. A low-cost muscle-computer interface was used to control wrist extension games with real-time sEMG feedback (TeleREINVENT).	Improvement in individual muscle activity and clinical outcomes. High adherence and enjoyment of the system.
Munoz-Novoa et al. (2024)	6 stroke patients with moderate to severe upper extremity (FMA-UE ≤ 50) using myoelectric pattern recognition combined with VR serious games. Data were collected through semi-structured interviews focusing on usability and user experience.	High motivation and engagement reported; challenges included electrode placement, system complexity, and need for assistance during home use.
Marin-Pardo et al. (2023)	4 chronic stroke participants with moderate to severe (FMA-UE = 19-40) completed 30 EMG-based VR telerehabilitation sessions over 6 weeks. Post-intervention semi-structured interviews were used to assess user acceptance and perceived functional change.	Increased use of affected arm in daily life; improved motivation; functional improvements were key driver for acceptance despite technical challenges.
Ito et al. (2024)	10 subacute stroke patients (FMA-UE = 10-65) performed daily EMG-controlled finger movement training using a wearable EMG sleeve integrated with a gamified exercise system over 7 non-consecutive days.	High enjoyment and motivation; gamification with adaptive difficulty helped maintain engagement during repetitive training.

Amin et al. (2025)	Randomized controlled trial with 52 subacute stroke patients (FMA-UE, control = 36.6 ± 7.4 , experimental = 40.84 ± 8.6) comparing VR EMG-based hand rehabilitation games plus conventional therapy vs conventional therapy alone over 4–9 weeks. EMG was used for movement classification and feedback.	Experimental group showed significantly greater improvements in hand function, dexterity, grip strength, and ROM with sustained follow-up effects.
Ito et al. (2023)	13 subacute stroke patients (FMA-UE = 10-66) used a multi-electrode EMG sleeve system controlling a serious game involving task-based upper limb movements.	High usability and enjoyment; participants rated EMG game more engaging than conventional exercises and showed willingness for continued use.
Dash and Lahiri (2020)	Mixed cohort of 12 stroke and 8 healthy participants completed VR-based grip training tasks controlled by sEMG signals from forearm muscles. Repeated sessions were used to assess strength and control changes.	Improved grip strength and task performance in stroke group; high acceptance of VR feedback-based rehabilitation.
Song et al. (2022a)	8 healthy subjects and 3 stroke patients trained using a gamified ADL-based rehabilitation system combining sEMG, FMG, and IMU sensors to classify functional hand and arm movements during task-based training.	Multimodal sensing (EMG + FMG + IMU) achieved highest accuracy (up to 95%); participants reported improved ability in daily hand activities
Song et al. (2022b)	10 stroke patients (FMA-UE 44.2 ± 13.9) used a multimodal sensing system (EMG, FMG, IMU) in two serious games designed to train 12 upper limb movements derived from FMA-UE.	Multimodal system improved classification accuracy (up to 81%) and increased user engagement and motivation
Ghassemi et al. (2019)	20 healthy participants trained with EMG-controlled cursor-based serious games over 3 sessions. EMG signals from forearm muscles were mapped into 2D control space using PCA.	Significant improvement in movement efficiency (67% faster completion); smoother and more targeted cursor control.
Donoso Brown et al. (2014)	9 chronic stroke patients underwent 4-week home-based EMG biofeedback training using a commercial game (Peggle) controlled by wrist flexor and extensor EMG signals.	Increased muscle activation and voluntary control; limited transfer to functional clinical outcome measures
Farasyfa et al. (2022)	Stroke patients trained wrist movements using a VR joystick system while EMG monitored muscle activity.	Improved wrist movement and muscle activation; VR enhanced motivation
Donosa Brown et al. (2015)	10 stroke patients completed a 4-week home EMG gaming program with interviews pre/post intervention.	Increased engagement, awareness, and use of affected limb in daily activities
Mejía et al. (2019)	Review of EMG and video game-based rehabilitation systems for upper limb stroke recovery.	EMG provides superior control and feedback; simple and intuitive games are essential
Seo et al. (2022)	20 chronic stroke patients (CMSA-H stage 4-5) completed an intervention consisting of 11 sessions (1 pre-test, 9 training, 1 post-test). sEMG from forearm and hand muscles (4 per limb) was used to control a cursor in 2D serious games. Training was performed unilaterally or bilaterally to improve voluntary activation of distal muscles in the paretic hand.	Significant improvement in movement efficiency: task completion time reduced by 46% and normalized path length reduced by 45%, indicating improved motor control and reduced unwanted muscle activation.

E Expert Interview

To gain insight into the clinical and practical aspects of hand rehabilitation after stroke and to better understand user needs within this context, this work employed a qualitative research design using a semi-structured expert interview. The interview contributed to the early-stage development of the design of the gamified hand rehabilitation system. The information gained from the expert interview, together with the findings presented in the literature [52, 53, 54, 37, 26, 31, 32, 33, 34, 35], was used to construct the user requirements of the system. The expert of the interview fulfilled the selection criteria in section E.1, and the semi-structured interview was followed the interview questions in section E.2. After the interview, the audio recording was transcribed as described in section E.3, at last, the interview was analyzed as described in section E.4. The information gained from the interview is found in section E.5

E.1 The expert selection criteria

A purposive sampling strategy was used in this work. Purpose sampling is recommended when selection information rich cases that can provide in-depth insights into a specific domain [121]. Given the exploratory design of this work, depth of information was prioritized over sample size, and one professional therapist with deep insight into the field of upper extremity rehabilitation was selected based on the following criteria:

- Clinical experience in neuro-rehabilitation
- Practical involvement in upper extremity training
- Experience with rehabilitation tools and assistive devices

The selected therapist has 40 years of experience in the neuro-rehabilitation field and meets the criteria above.

E.2 The interview questions

The interview was performed in danish, below follows the interview question in Danish and translated to English in Table 16.

E.3 Transcription guide

To transcribe the interview, the audio file was uploaded as a .m4a file in **Microsoft Word** where the **Transcribe** option was used. After the automatic transcription, the transcription was corrected while listening to the audio file based on the transcription criteria:

- The transcription does not include filler words and phrases such as "*Øhh*", "*Hvad hedder det nu..*", etc.
- Possible transcription errors were corrected.
- Pressing on words and exclamations were not highlighted.

E.4 The Analysis of the interview

The interview data were analyzed using thematic Qualitative Content Analysis (QCA) which is one of the most popular methods for analyzing qualitative data, and is a proven method in qualitative research [122]. The analysis process was performed in five iterative stages proposed in the work by Udon Kuckartz [122]. The five steps in the iterative circle were performed as:

Nr.	Danish	English
1	Hvilke fokuspunkter er typiske for genoptræning af hele overekstremiteten, inklusive hånd, håndled og fingre?	What are the typical focus points for upper limb rehabilitation, including the hand, wrist and fingers?
2	Hvilke kognitive begrænsninger og hvordan tages der højde for de kognitive begrænsninger under genoptræningsøvelserne for overekstremiteten?	What cognitive limitations and how are cognitive limitations taken into account during upper limb rehabilitation exercises?
3	Har I specifikke øvelser målrettet hånd/fingre/håndled og hvordan adskiller de sig fra træning af resten af armen?	Do you have specific exercises targeting the hand/fingers/wrist and how do they differ from training the rest of the arm?
4	Hvilke træningsredskaber bruger I i dag til overekstremitetstræning?	What training tools do you currently use for upper limb training?
5	Hvilke tiltag benytter I jer af for at gøre træningen mere motiverende for patienterne?	What initiatives do you use to make the training more motivating for patients?
6	Er der anden information du vil supplere med, som du vurderer vil være nyttigt for os at tage i betragtning i vores designproces?	Is there any other information you would like to add that you think would be useful for us to consider in our design process?

Table 16: Interview questions in Danish and English.

1. Prepare and reading the data intensively: The interview was transcribed by following the Transcription Guide in section E.3, afterwards the interview was read through intensively multiple time to ensure familiarization with the data.
2. Building the coding frame: In the first iteration of this step the main categories where created, by mixing the driven inductive method where the main categories were created from the research questions and the data driven inductive method where additional main categories where made based in the content of the interview data. In the next iterations the data driven method were used to create the subcategories. The main categories and subcategories can be seen in table 17.
3. Coding the data: In the first iteration the data were coded into the created main categories, and in the next iterations into to the subcategories. The iteration of creating sub categories and coding the data into the subcategories stopped when each subcategory contained only one short main statement.
4. Analyzing the coded data: To analyze the coded data each short statement was written out to more meaningful sentences, that captured the essence if the data segments, some main points where closely connected to each other and where therefore combined into small text segments.
5. Presenting the results: To present the result the small text segments where collected, and send to the interviewed therapist, to ensure factual correctness in the analysis. The therapist was asked to leave comments in the was phrases and/or interpretations that where not correctly understood. Lastly the analyze where adjusted according to the therapists comments. The results of the analyze can be seen in section E.5.

Main theme	Sub Themes
Exercises	Exercises shall mimic everyday activities, Therapy Tools, Sensibility, Thumb, Edge position of the arm, Arm strength, Patient Progression, Dose and Quality of the exercises
Cognitive limitations	Memory and attention constraints, Aphasia
Patient independence	Responsibility for own training, Self-training
Motivation	Motivation
Missing	Missing feedback, Missing dosis

Table 17: Main and subcategories developed in the second step of the iterative qualitative content analysis circle.

E.5 Findings from the Interview

The findings of the interview was listed below:

- The goals and the sub goals of the rehabilitation are planned in close collaboration between patient and therapist, where it is important that the exercises reflect everyday activities in order to enable the patient to resume daily life again. To analyze which components to focus on in the training, observation, and analysis of the movement *functional reach* is used, among other things.
- During rehabilitation, there is a big focus on all the complex functions of the hand and the thumb. It is experience that many struggle to get the thumb activated.
- Many patients are challenged by the edge position of the arm, but this functionality is important for reaching and grasping objects.
- Rehabilitation tools for exercises contribute to making the exercises more understandable and providing real-time feedback, while rehabilitation tools can be directly related to everyday activities.
- Dosage is central to promoting neuroplasticity, but it is important not to compromise on the quality of the exercises. It has been experienced that gamification can strengthen the motivation, but the quality in the form of precision, selectivity, and relevant use of the arm, disappeared.
- Almost all patients have cognitive disorders, it is therefore important that the training is adapted to the patient's cognitive constraints. The typical cognitive challenges are limited to memory span and attention disorder. The tasks must therefore be simple and intuitive to understand, it is important that the exercises are not too complicated.
- Patients are guided to take responsibility for their own rehabilitation as it is part of their new life, where the therapists act as training coaches where they, among other things, guide the patients to document their training in their personal training folder. However, the patients only receive the responsibility that is assessed as possible based on the patients cognitive level.
- Ongoing feedback on the exercises will help maintain motivation, but there must still be a focus on the quality.
- It is difficult to achieve the optimal amount of training because patients the cognitive challenges need nudging to complete the training and there are not enough therapists to do so. It is therefore crucial to keep the patient motivated during the training. Feedback on the dose amount and correctness of the exercises will help maintain motivation and focus on the training session.

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